

模拟电子技术基础

Analog Device

2.1 Bipolar Junction Transistor

晶体三极管

陈晓龙 Chen, Xiaolong

Department of Electrical and Electronic Engineering

2.0 History of transistor

2nd generation electronic device-- **transistor** 晶体管

1947, Bell Lab



Transistor is made of semiconductors(半导体)

John Bardeen (1908-1991), Walter House Brattain(1902-1987), and William Shockley (1910-1989)

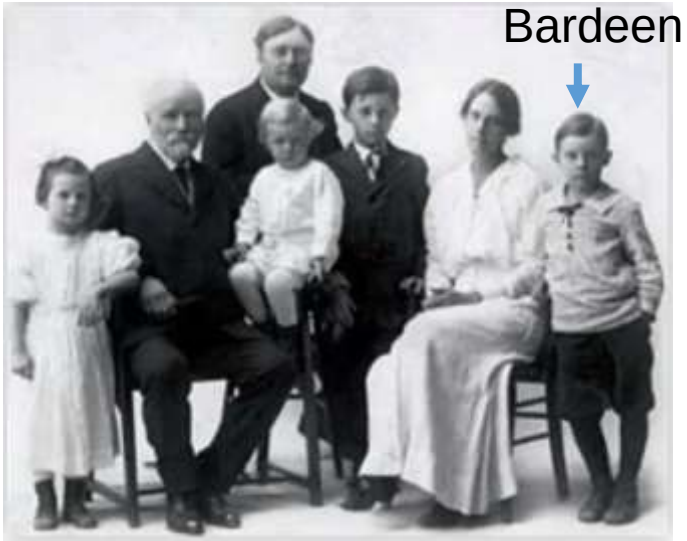
John Bardeen



约翰·巴丁(John Bardeen)

He won Nobel Physics Prize twice

- 1956 for inventing transistors (Bardeen, Shockley, Brattain)
- 1972 for the BCS superconducting theory (Bardeen, Cooper, Schrieffer)



- Born in 1908.05.28
- Went to University of Wisconsin at 15 years old
- Went to Princeton University

Majoring in Electrical Engineering

Study Solid Physics

Get married!

- Join Bell lab in 1945

Work with Shockley and Brattain

Aim at developing next-generation transistor (Shockley is the team leader)



巴丁与珍妮

- Shockley proposed the metal-semiconductor junction structure

No current rectification effect are observed

- Bardeen proposed the surface states in semiconductors



The first transistor in the world

Shockley was not listed in the inventor list



- Shockley invented junction FET secretly

Kick out Bardeen and Brattain from Bell lab

Left Bell lab in 1955

- Open his own lab in Silicon Valley in 1956

Attract many young scholars:

Robert Norton Noyce, left in 1957

Gordon Moore, left in 1957

- Noyce, Moore, et al. founded Fairchild semiconductor in 1957

- Noyce and Moore left Fairchild and founded Intel



Noyce



Moore

- 9 years after the invention of transistor, Bardeen, Shockley and Brattain 1956 won Nobel Physics Prize.

有天晚上，一班物理学家和夫人自发组织拿着手电筒当火炬高歌祝贺巴丁，巴丁赶紧制止他们，“你们轻一点，别吵到我太太睡觉。”巴丁的“妻管严”（“henpecked”）是出了名的，连报纸的头版头条都是：“本地一女子的丈夫获得诺贝尔奖”。无论实验室工作有多忙，家里饭菜都是巴丁一个人包揽。



- In 1957, Bardeen, Cooper and Schrieffer proposed the BCS superconducting theory.
- In 1972 Bardeen, Cooper, and Schrieffer won Nobel Physics Prize for the BCS superconducting theory.

“Bardeen Number 巴丁数”



约翰·巴丁(John Bardeen)

The only person ever to win two Nobel Prizes in physics, John Bardeen's name is barely known outside of the physics community.

Bardeen Number: the ratio of a physicist's substance to his self-advertisement.

Transistor

Advantages:

- a. Small & light
- b. Long lifespan
- c. Low power consumption

Disadvantages:

- c. Sensitive to temperature
- d. Poor overload capacity



2.1 Bipolar junction transistor (BJT)

Transistor is short for “Bipolar junction transistor”

双极型晶体管， 又称半导体三极管， 简称为晶体管

The **amplification** and **switching** functions of transistors have promoted the leap of electronic technology



¥2.50

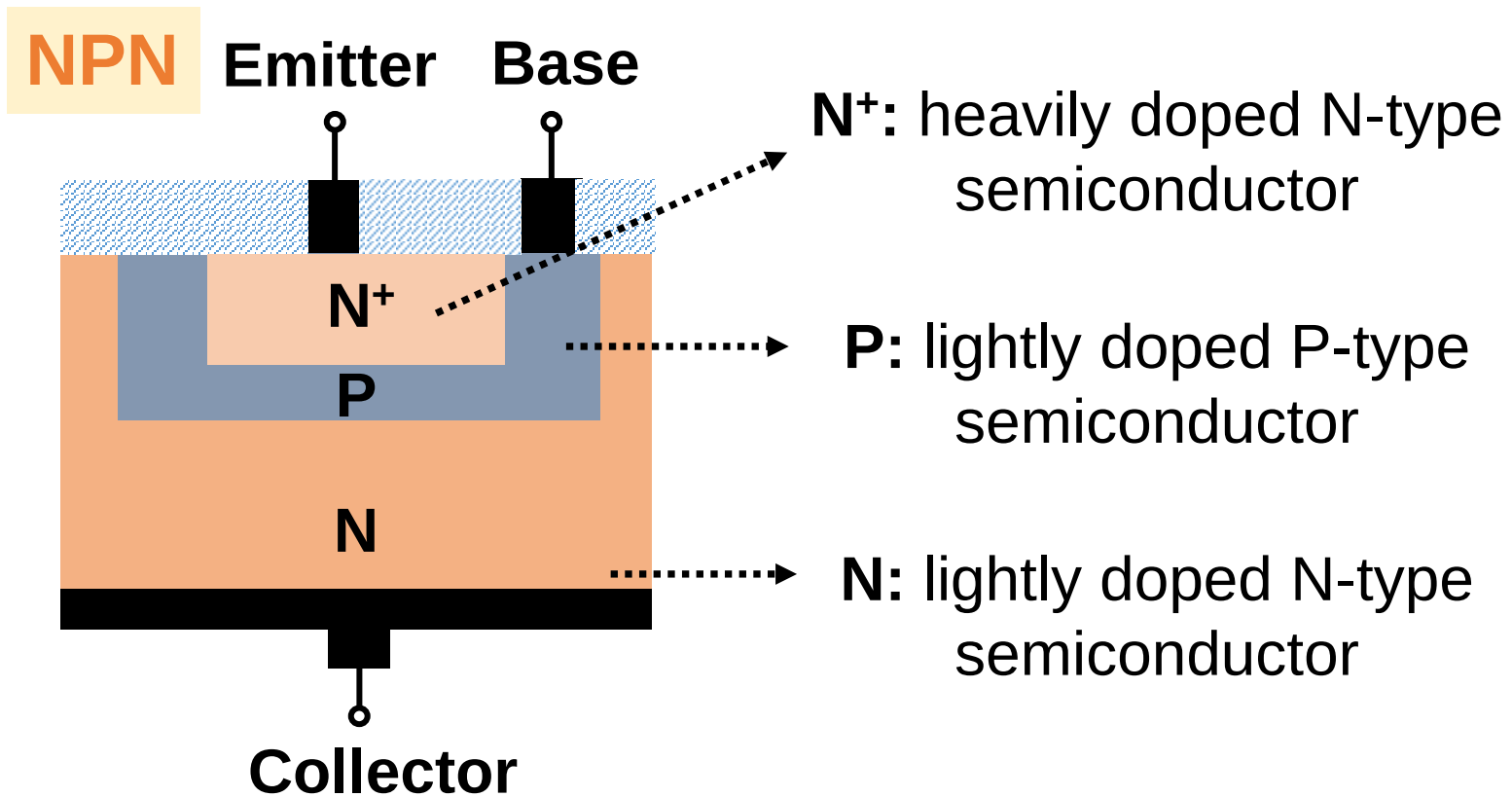
NPN硅 (10个)

One of the key function of transistor is **amplification**

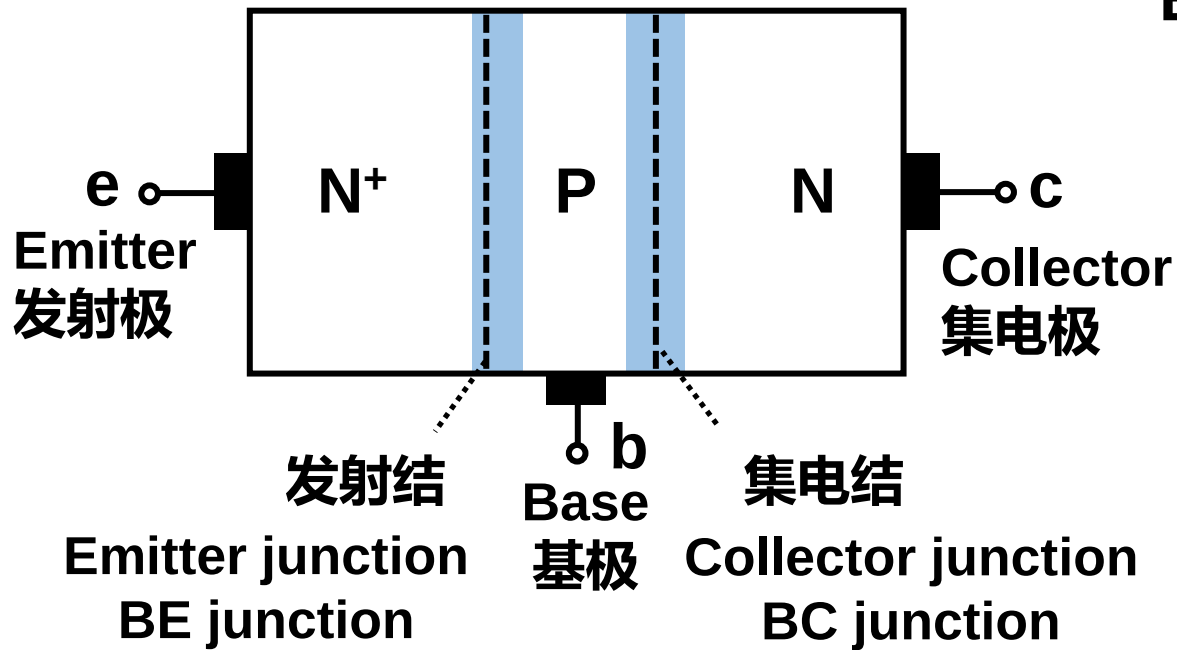


2.1.1 Structure of BJT

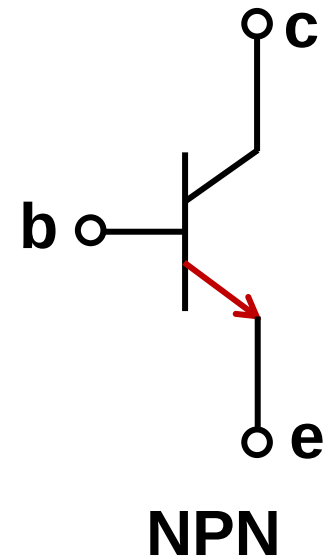
Transistor configuration: **NPN** or **PNP** -type



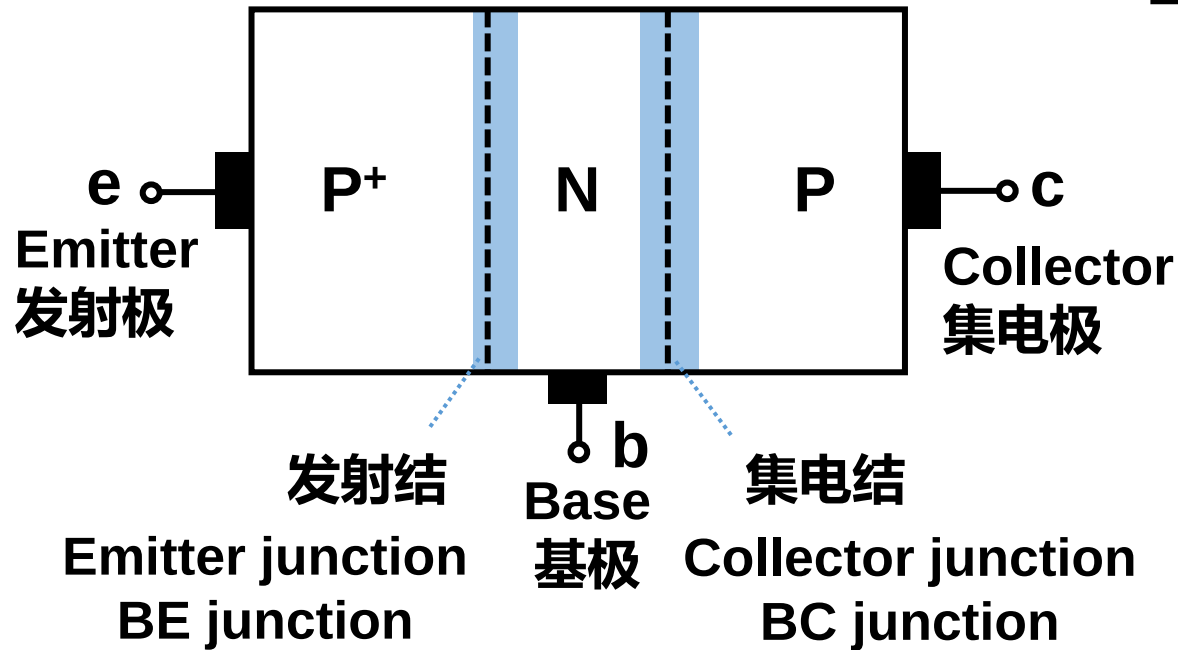
NPN transistor



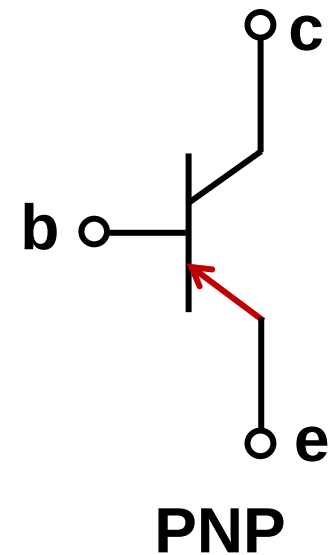
Electronic symbol

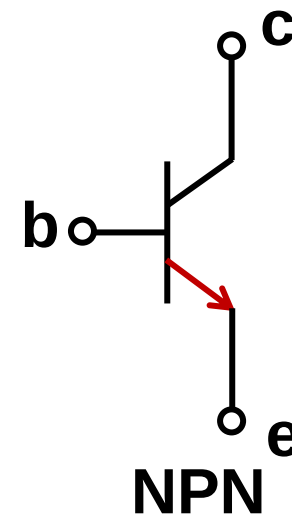
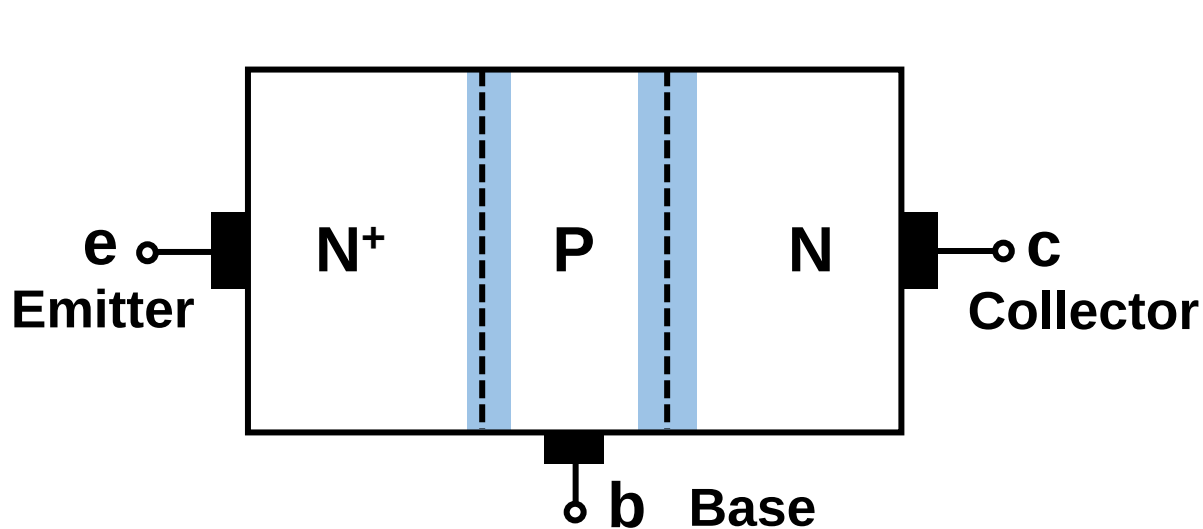


PNP transistor



Electronic symbol

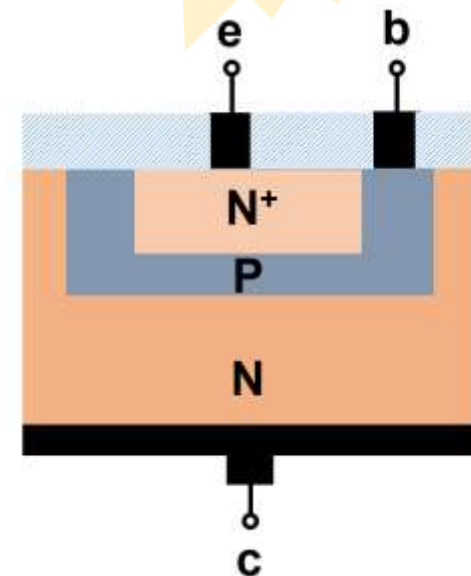




Key features of transistor (for amplification)

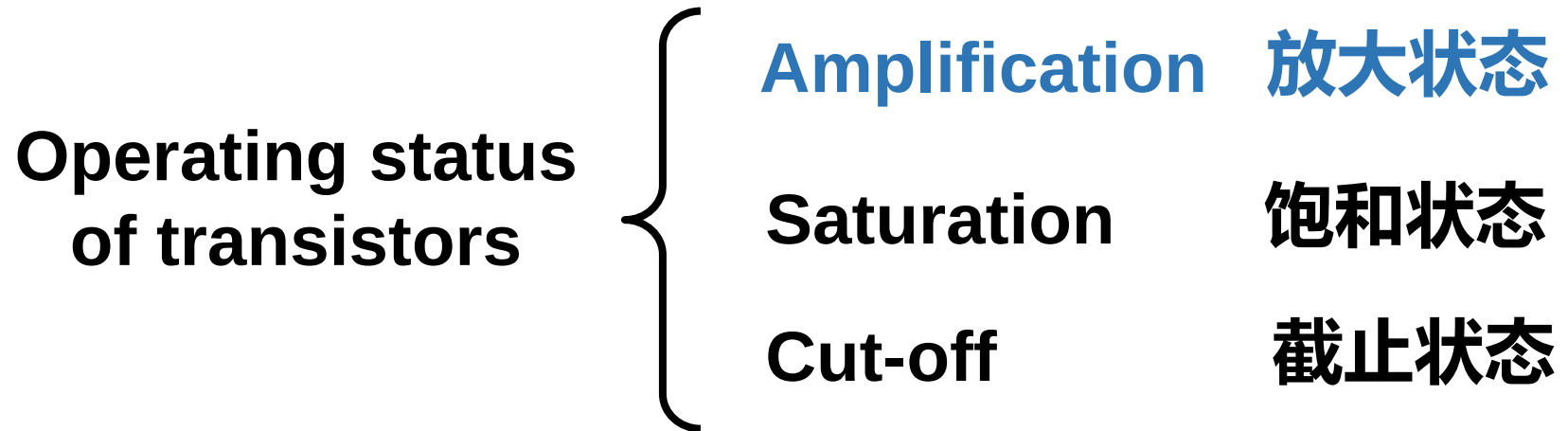
- (1) Emitter region: high doping, small area
发射区掺杂浓度大，面积小。
- (2) Collector region: low doping, large area
集电区掺杂浓度低，集电结面积大。
- (3) Base region: low doping, very thin
基区掺杂浓度很低，且很薄。

Why?

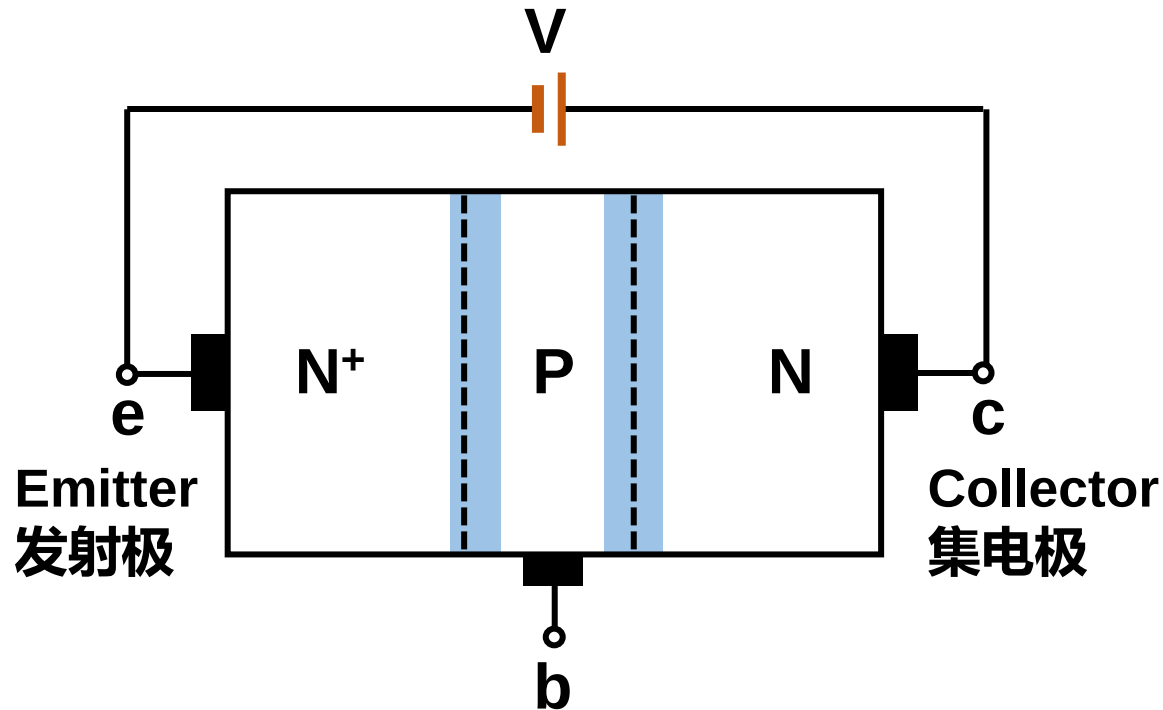


2.1.2 Working principle of BJT (NPN)

Take **NPN-type** transistor as an example



Two-terminal connection



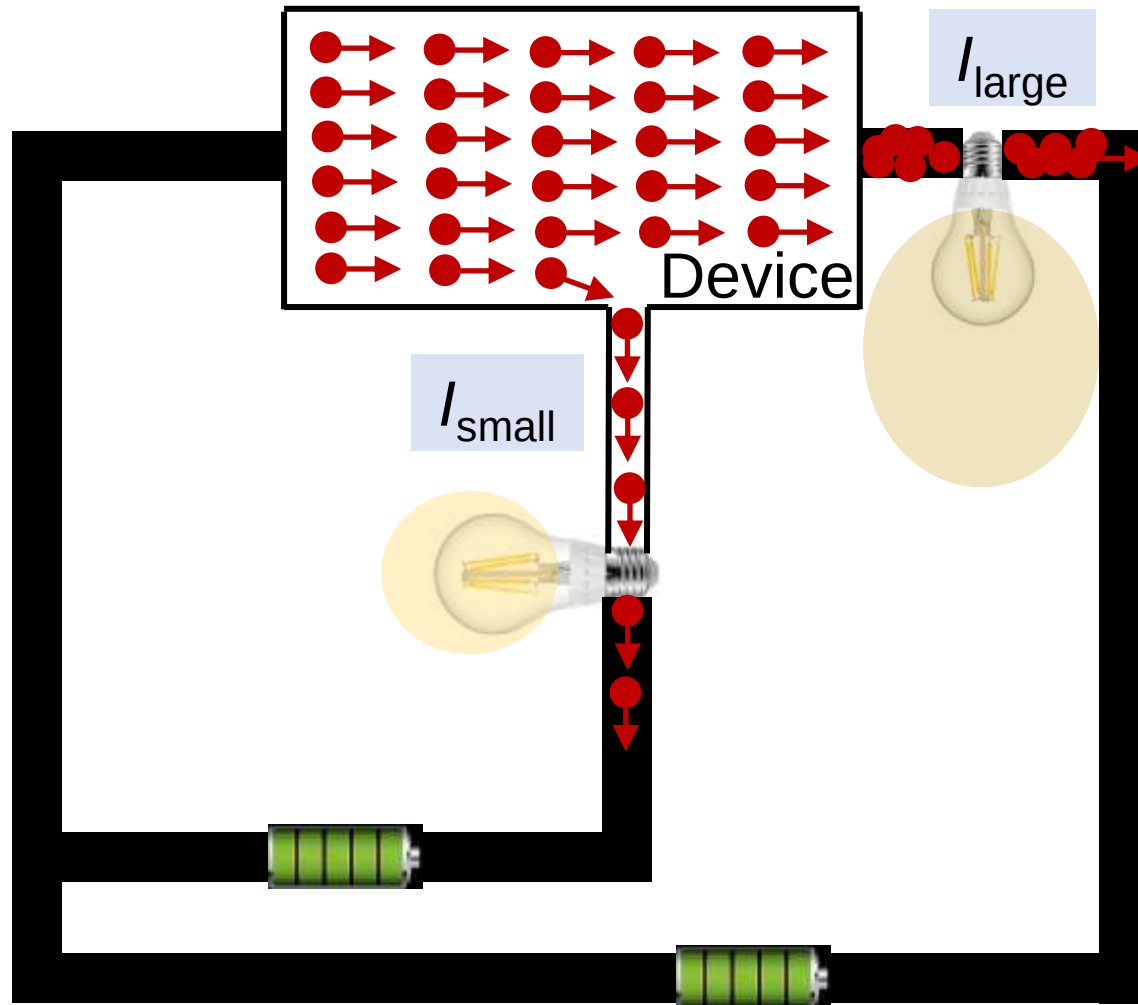
One PN junction is always reverse biased!

How to realize amplification?

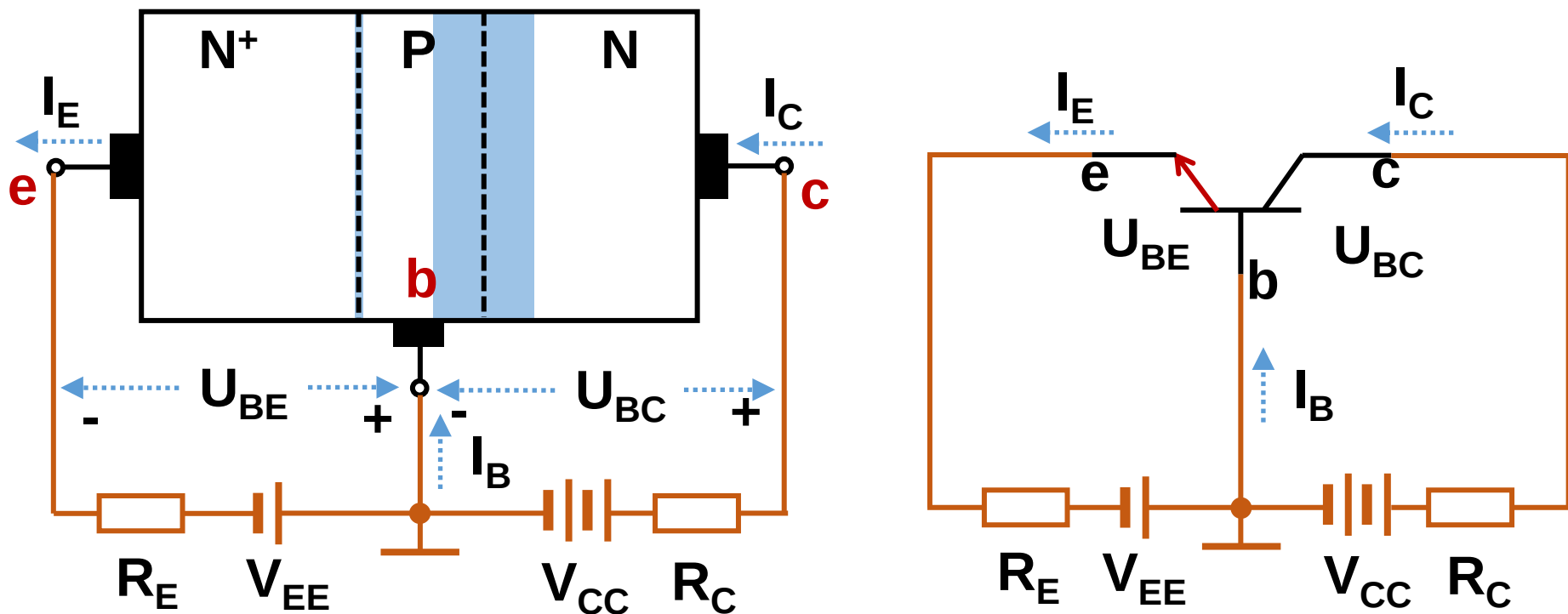
Current amplification: $I_{\text{large}} \propto I_{\text{small}}$

or

Power amplification: $P_{\text{large}} \propto P_{\text{small}}$



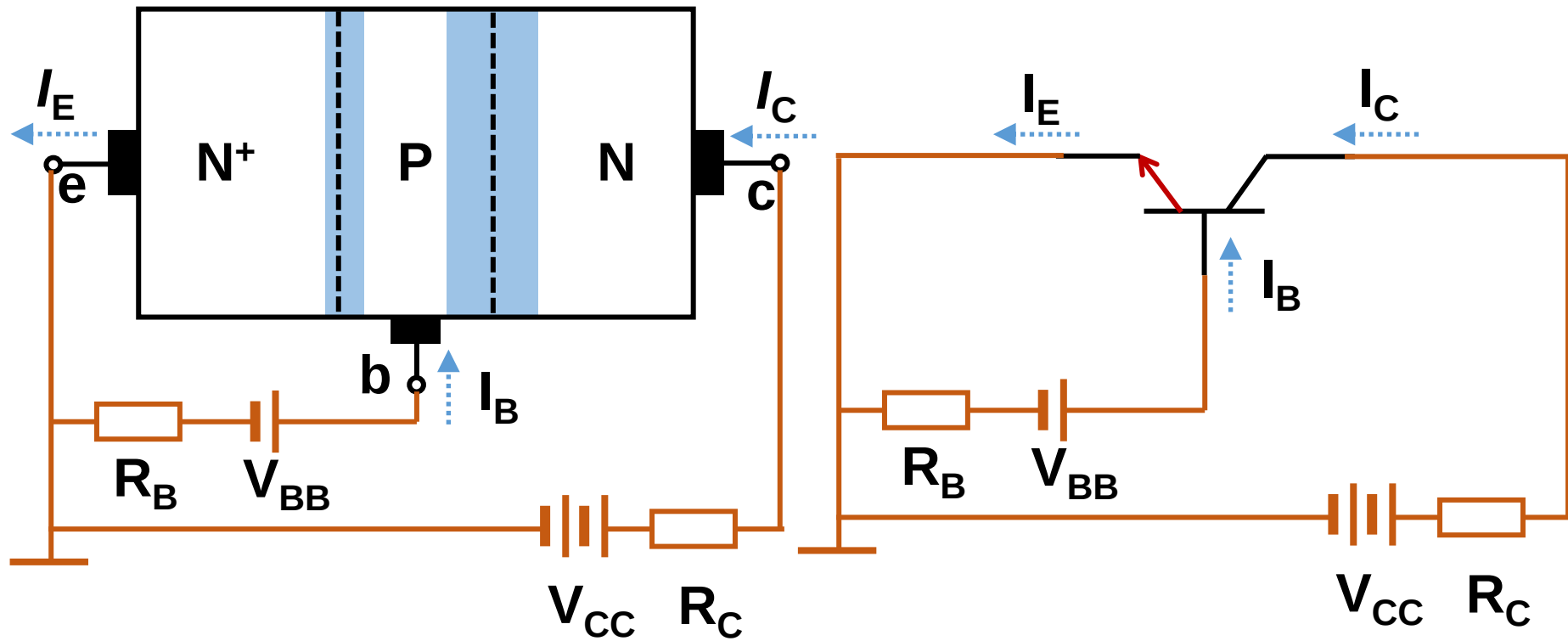
Amplification mode: BE junction forward bias & BC junction reverse bias 发射结正向偏置, 集电结反向偏置



Common-**base** amplifier circuit

共基极放大电路

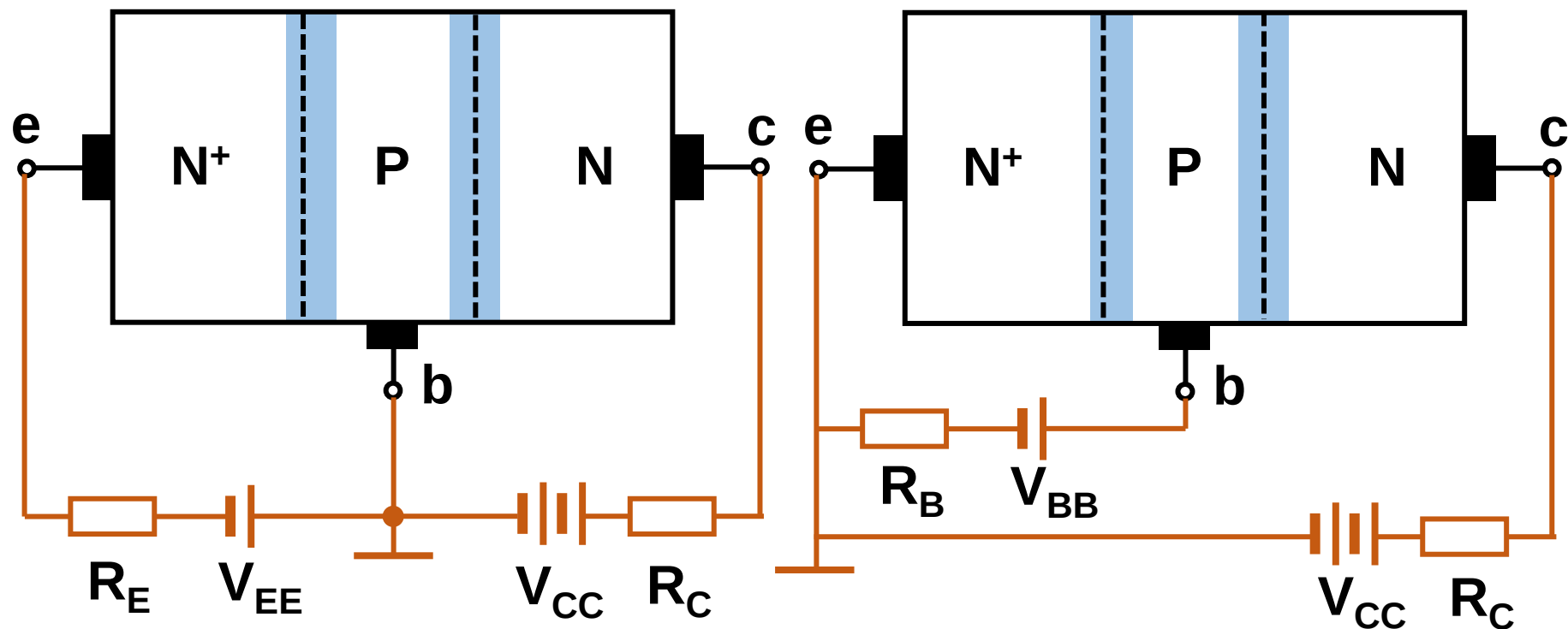
Amplification mode: BE junction forward bias & BC junction reverse bias



Common-emitter amplifier circuit

共射极放大电路

Amplification mode: BE junction forward bias & BC junction reverse bias **放大模式:** 发射结正向偏置, 集电结反向偏置



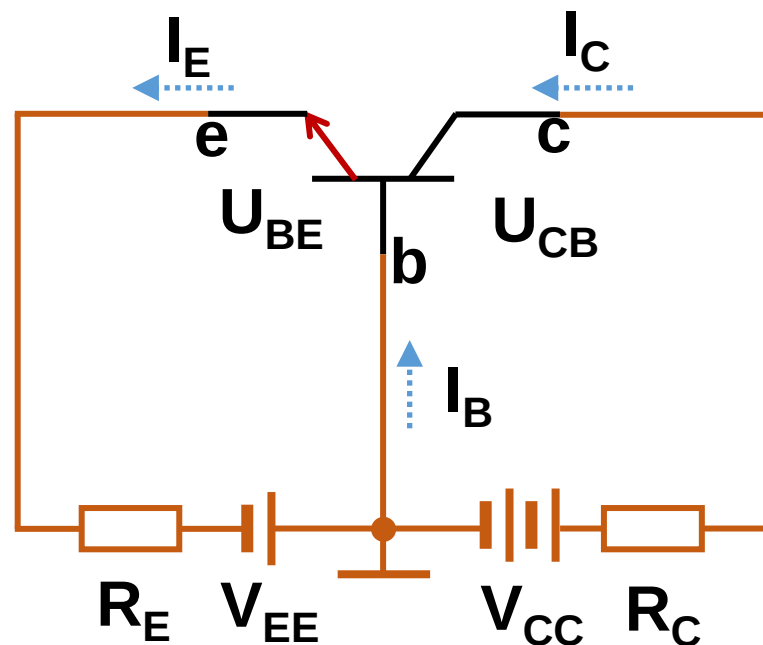
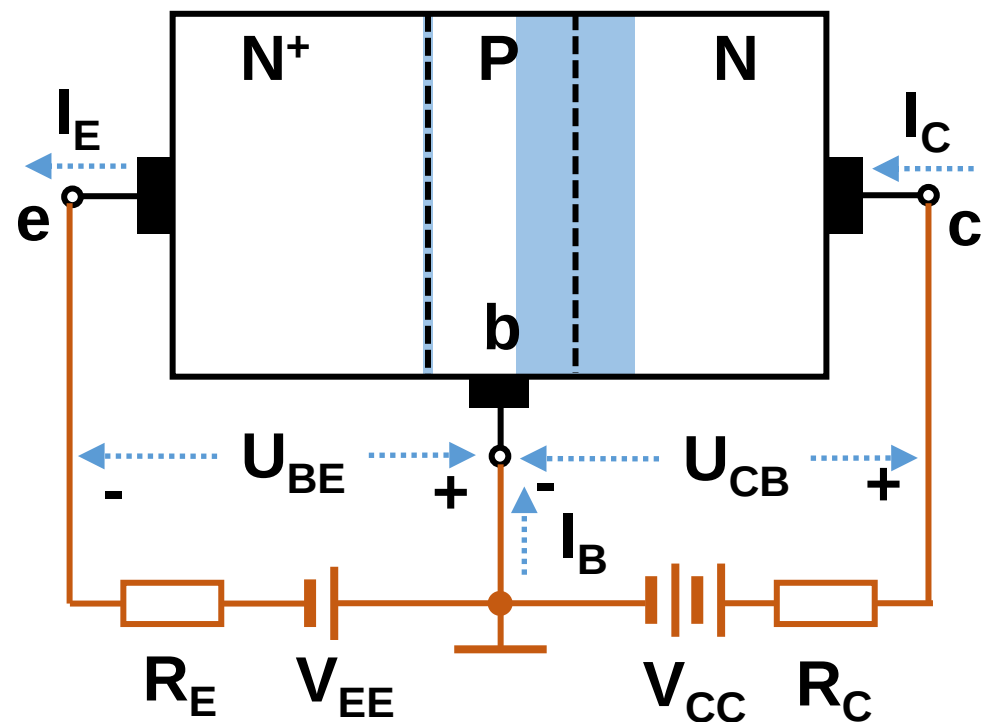
Common-base amplification circuit Common-emitter amplification circuit

共基极放大电路

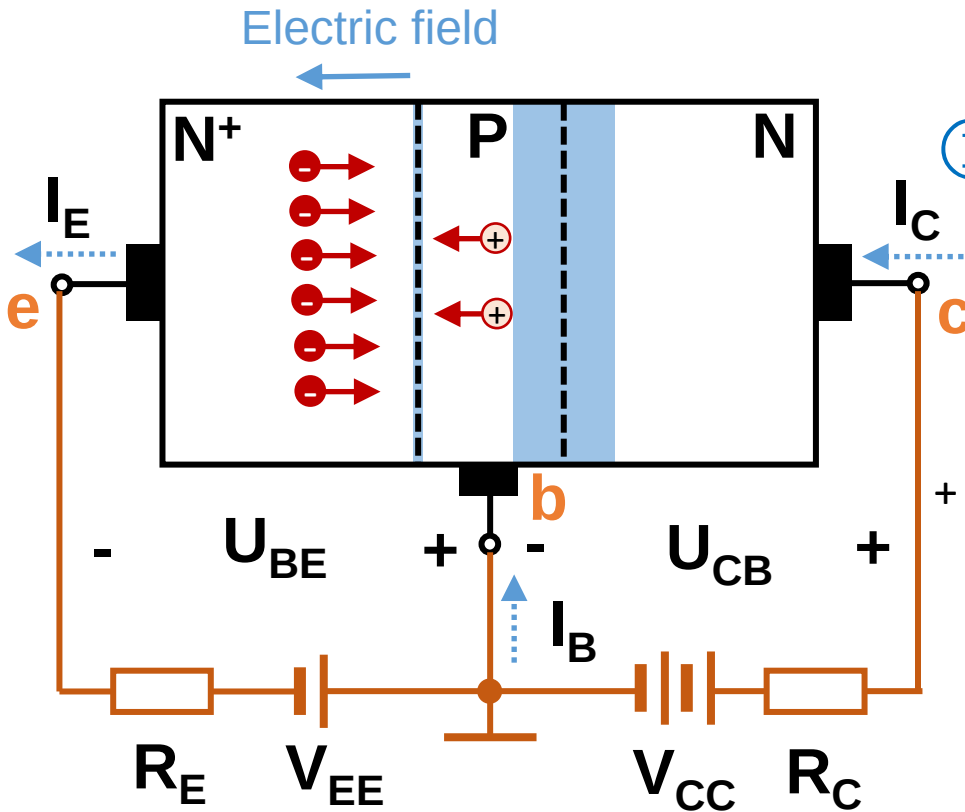
共射极放大电路

Common-base amplification circuit

共基极放大电路



Forward bias of BE junction

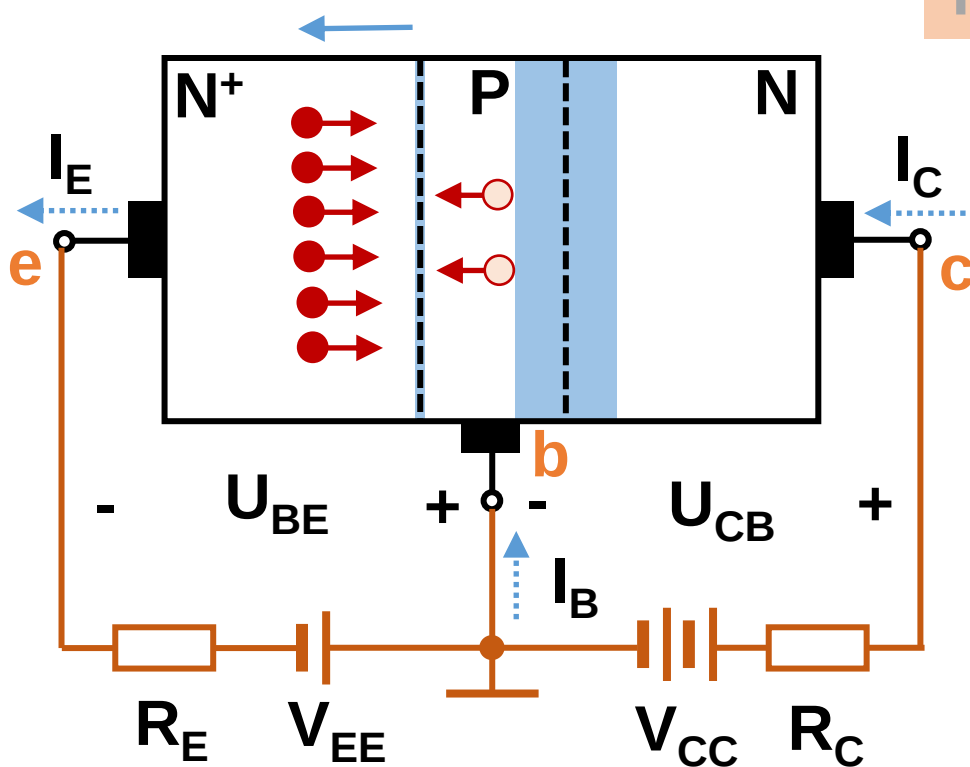


① BE junction becomes thin.

BE junction: The electron concentration in N^+ region is much higher than the hole concentration in P region.

② Electrons (major carrier) in N^+ region drift to P region under electric field.

Electric field



Forward bias of BE junction

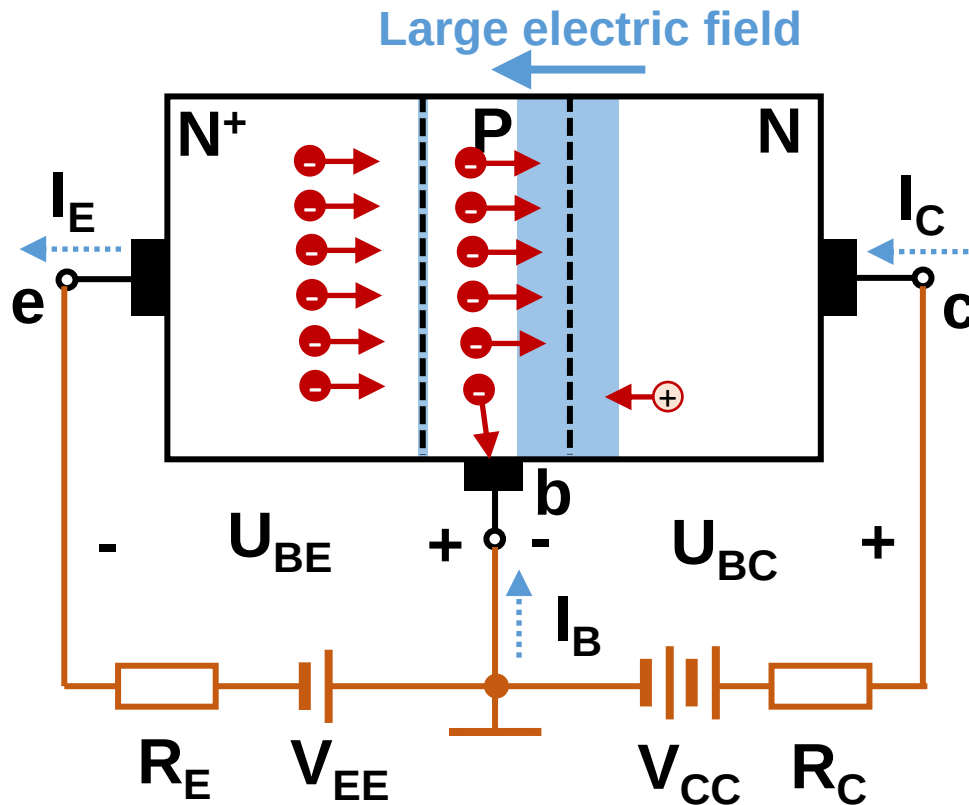
BE间的PN结宽度变薄

多数载流子导电为主;

BE结: N^+ 区是高掺杂, P 区是低掺杂, N^+ 区电子浓度远大于 P 区空穴浓度, 以 N^+ 区电子导电为主, 空穴忽略不计

N^+ 区的大量电子在外电场的作用下漂移到 P 区

Reverse bias of BC junction



③ BC junction becomes thicker.

Internal electric field becomes stronger.

④ Most electrons are accelerated by the electric field of BC junction and collected by the **collector**. Only small portion is collected by **base**.

$$I_C \gg I_B$$

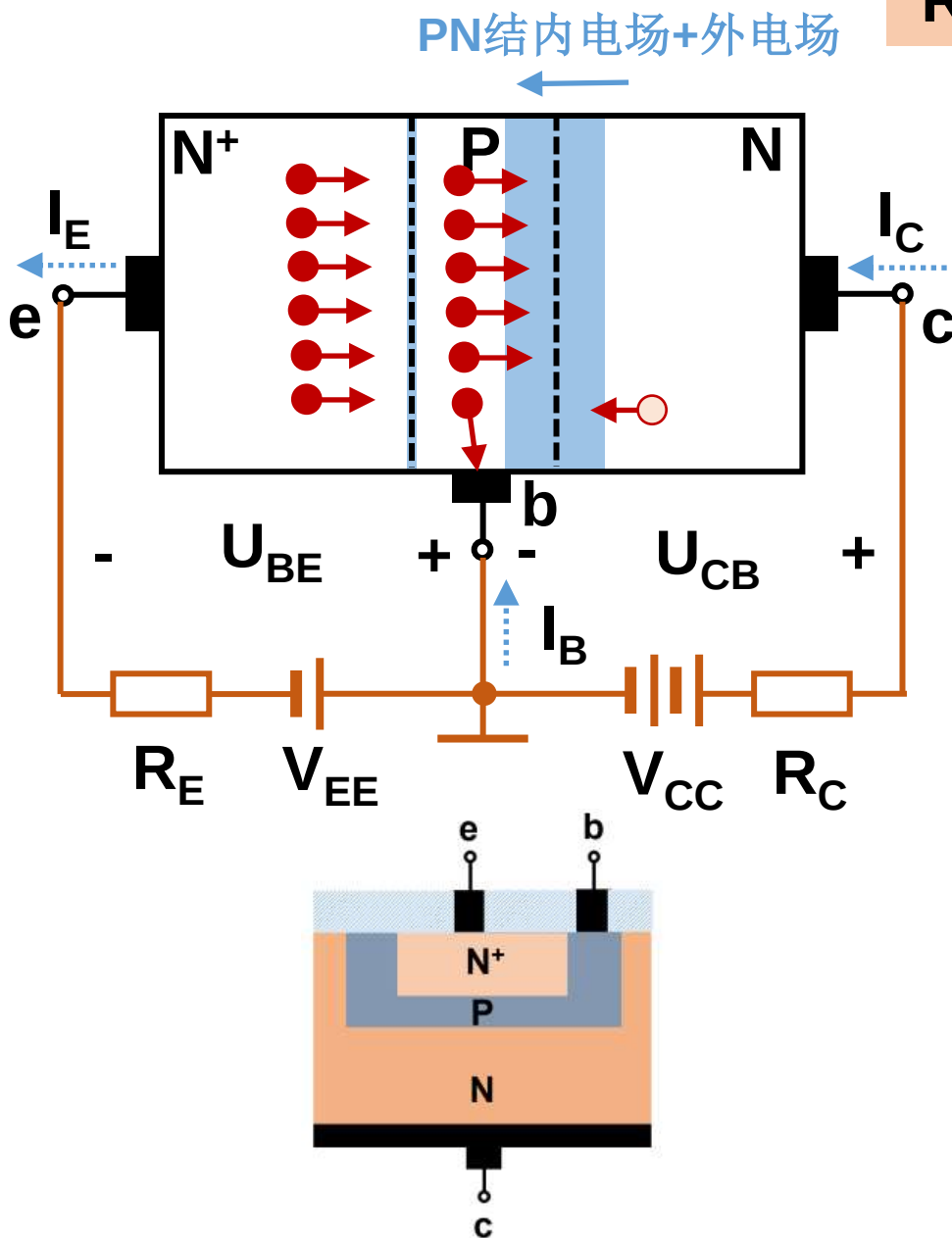
Reverse bias of BC junction

BC间的PN结宽度变厚

少数载流子导电为主；

BC结： N区和P区都是低掺杂，电子和空穴浓度都很低，远小于从N⁺区漂移过来的电子浓度

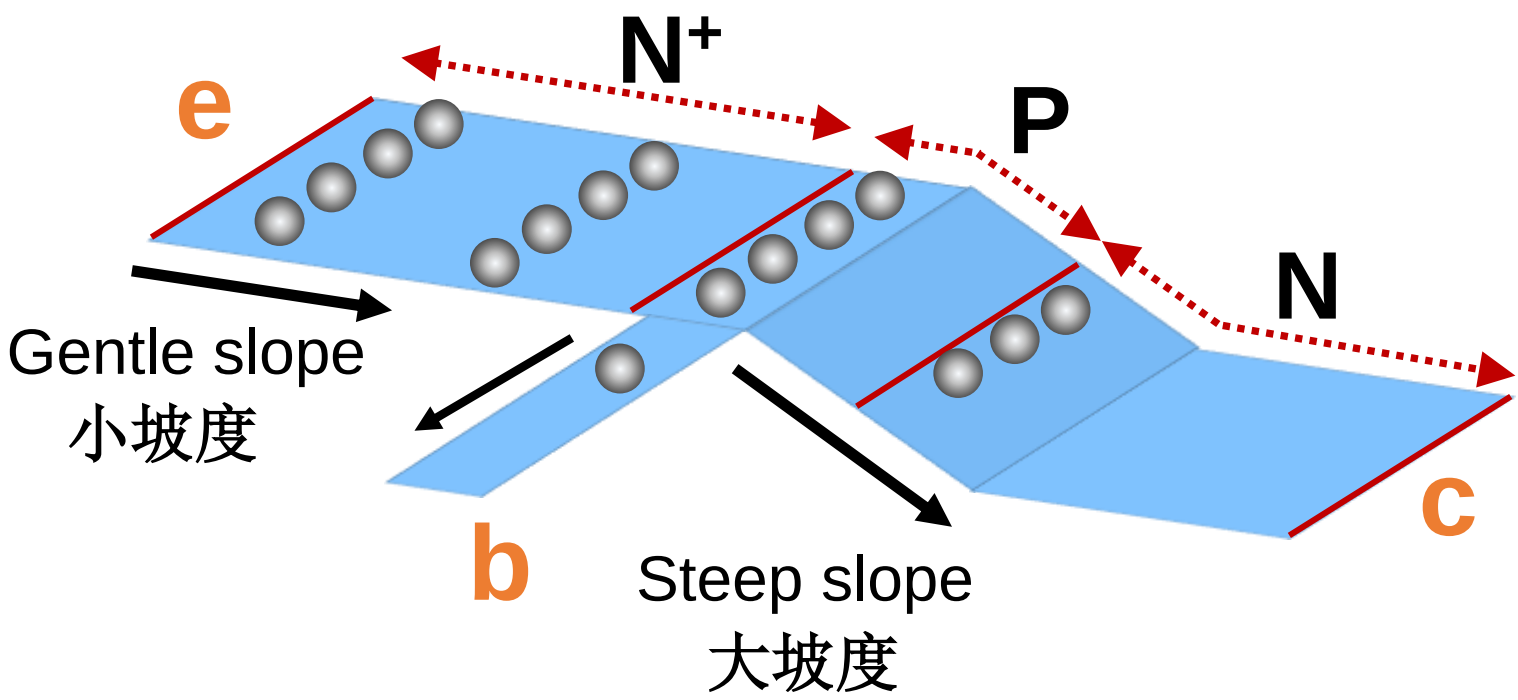
从N⁺区漂移过来的电子在BC间的内点场和外电场作用下，大部分漂移到N区形成电流 I_C ，少部分被基极收集形成电流 I_B 。还有少部分电子与P区的空穴复合，但是可以忽略不计



Treat electrons as footballs

把电子等效成皮球

Treat electric fields as slope of hills 电场强度等效为山坡的坡度



Treat electrons as water molecules

Emitter

The source of a river



Base

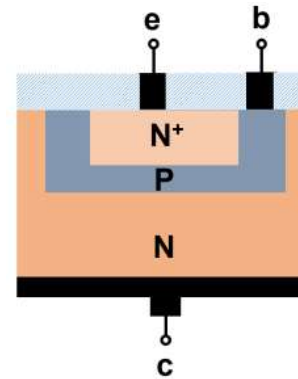
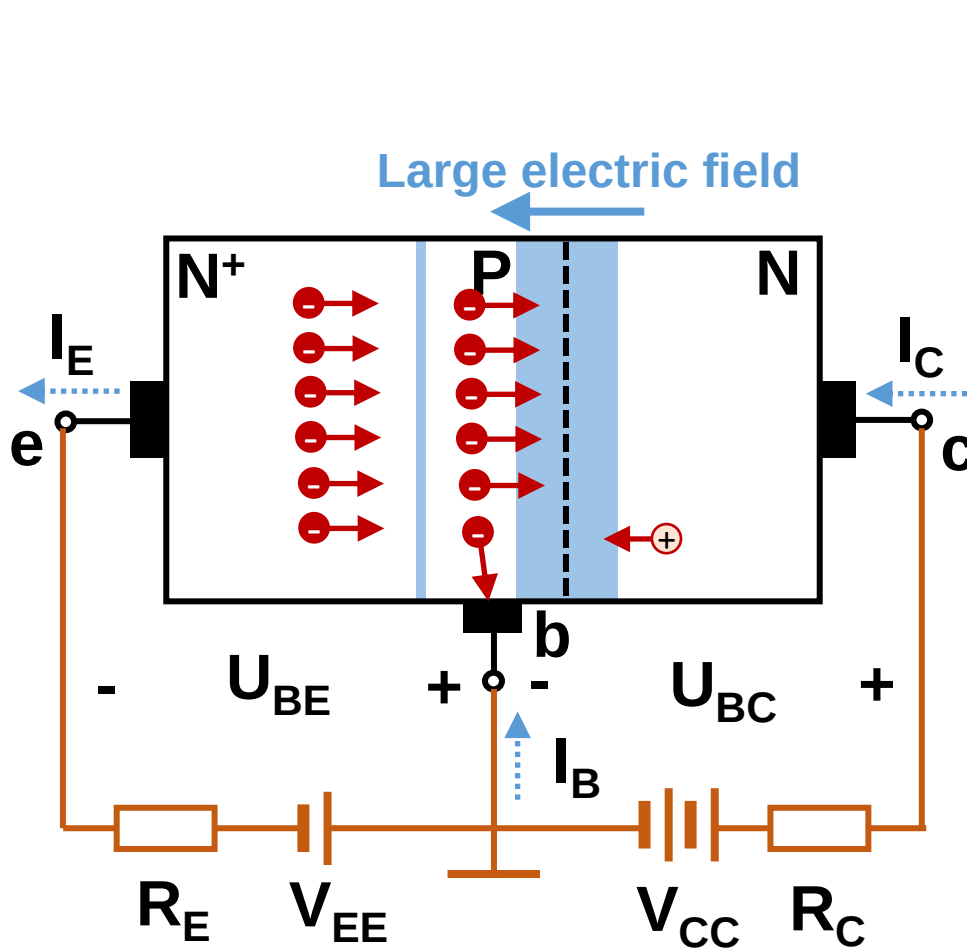
I_B



Collector

I_C

$$I_C \gg I_B$$



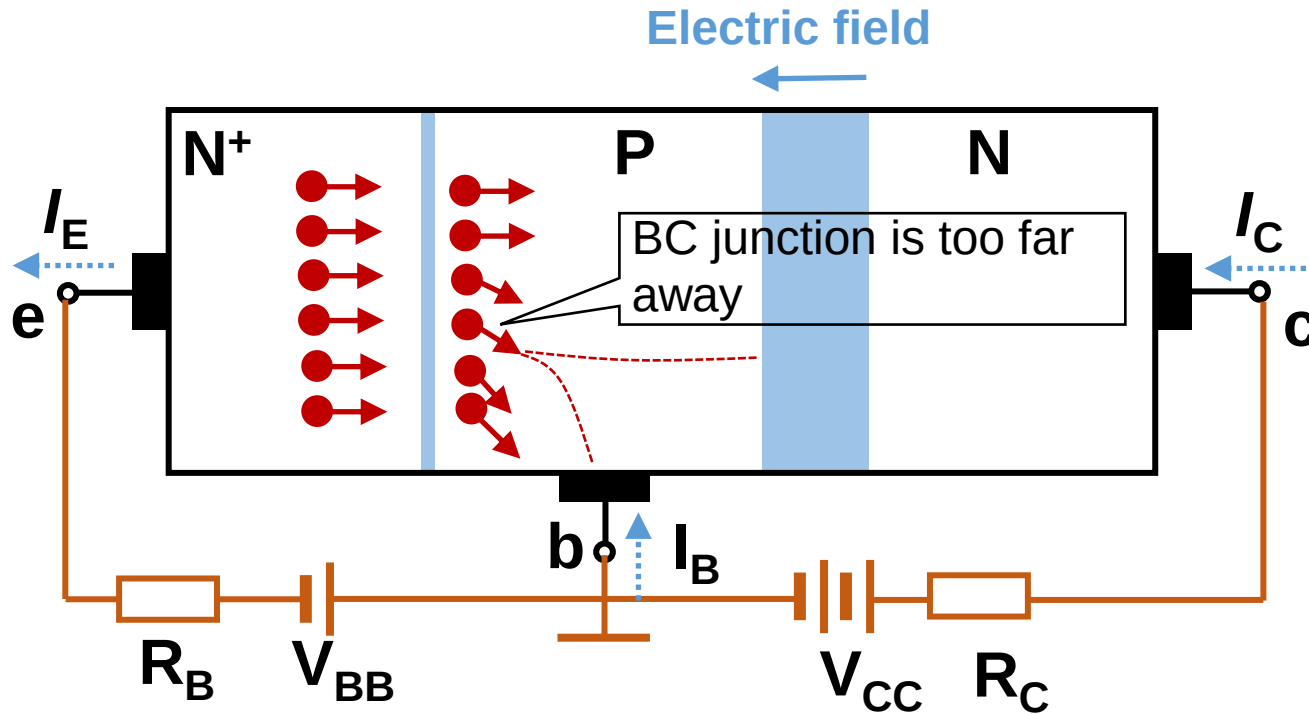
Question!

Why N^+ region should be highly doped (1)?

Why P region should be thin? (1)

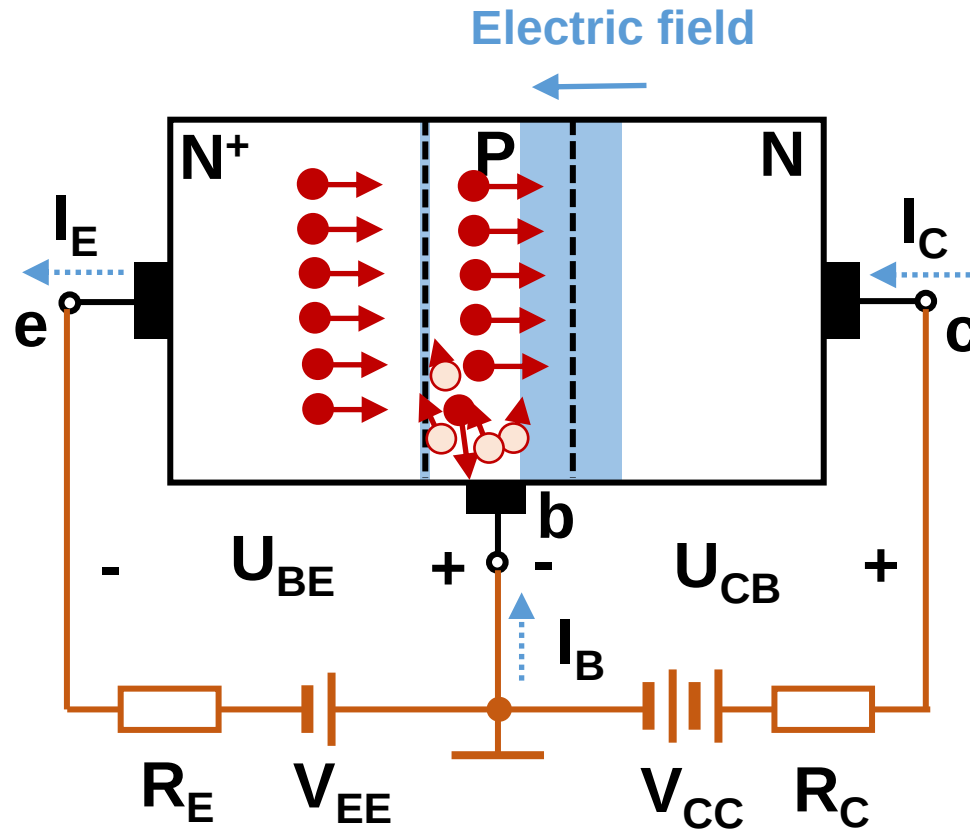
Why P region should be lightly doped? (1)

Assume P-region was very thick

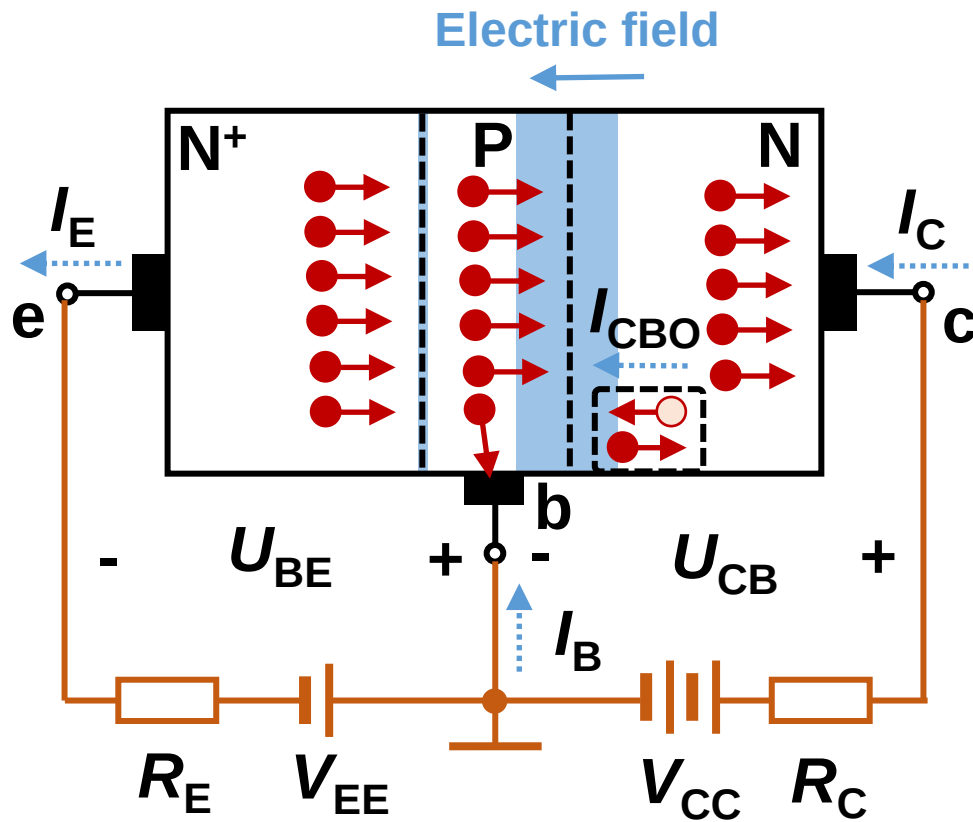


I_B increases significantly! I_C/I_B decreases!

Assume P-region was heavily doped



I_B increases significantly! I_C/I_B decreases!



I_{CBO} : the reverse saturation current when emitter is open.

$$I_E = I_C + I_B$$

$$I_E \approx I_C \gg I_B$$

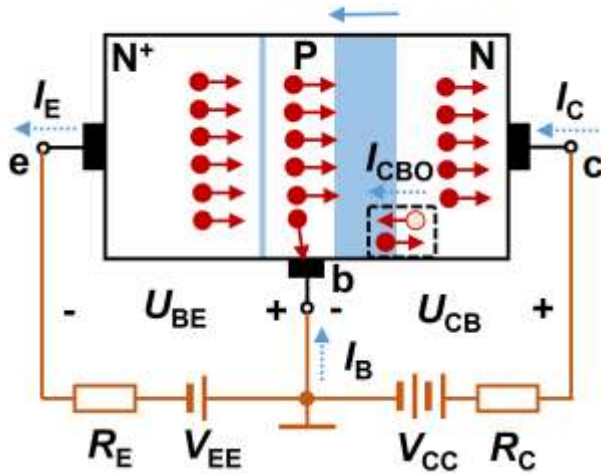
**Common-b DC
amplification factor**
共基极直流放大系数

$$\bar{\alpha} = \frac{I_C - I_{CBO}}{I_E} \approx \frac{I_C}{I_E} < 1$$

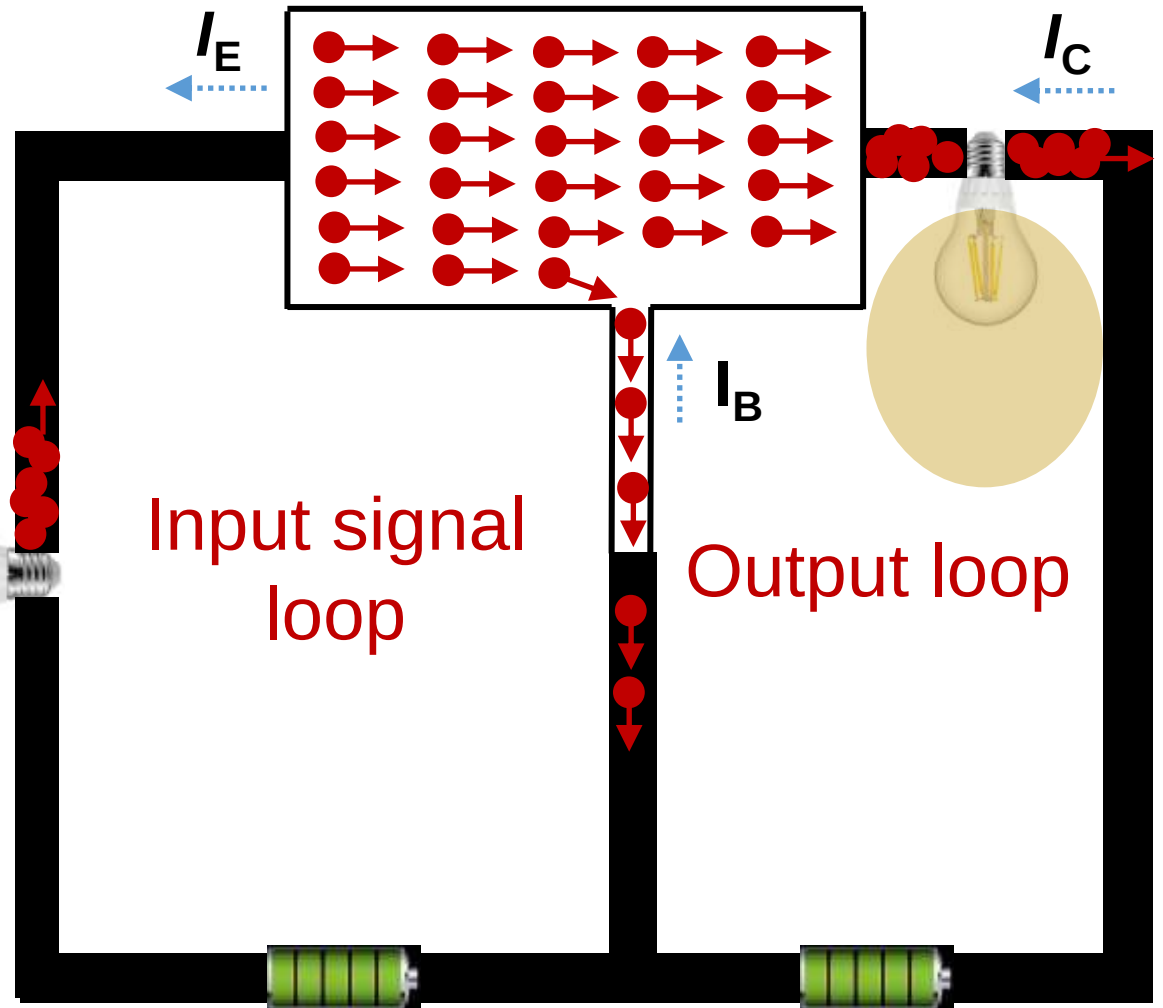
Q: Why common-b amplification coefficient is not defined as:

$$\bar{\beta} = \frac{I_C}{I_B} \gg 1?$$

Common-base amplifier

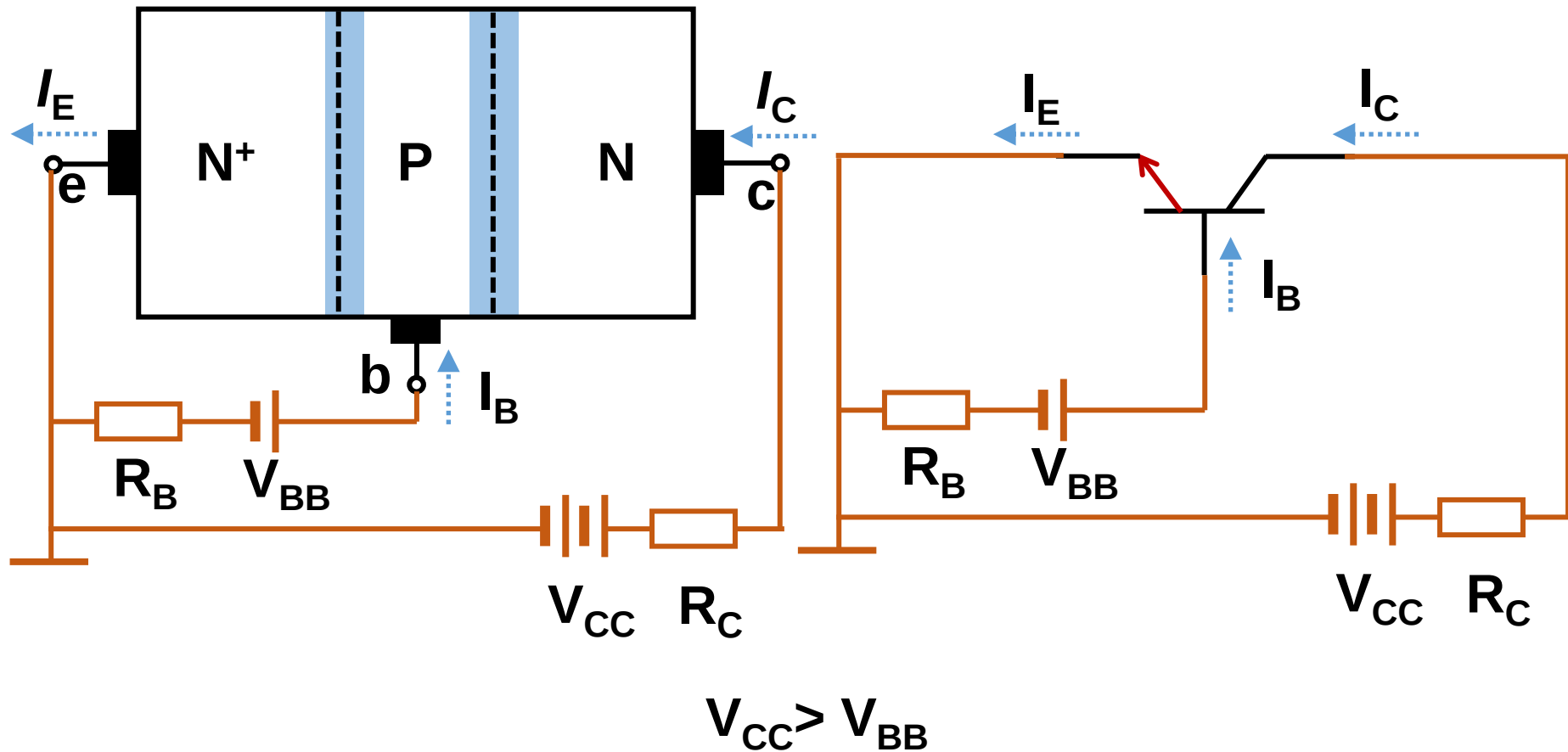


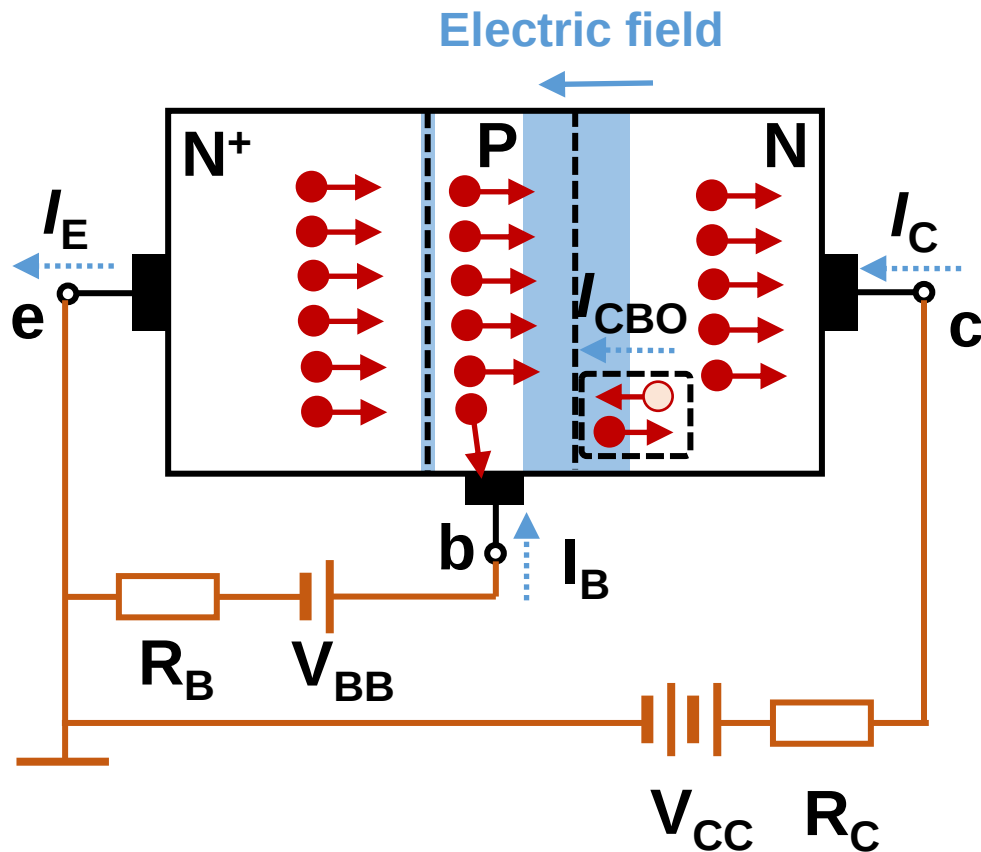
$$\bar{\alpha} = \frac{I_C}{I_E} < 1$$



2.1.4 Common-emitter amplification circuit

共射极放大电路





$$I_E = I_C + I_B$$

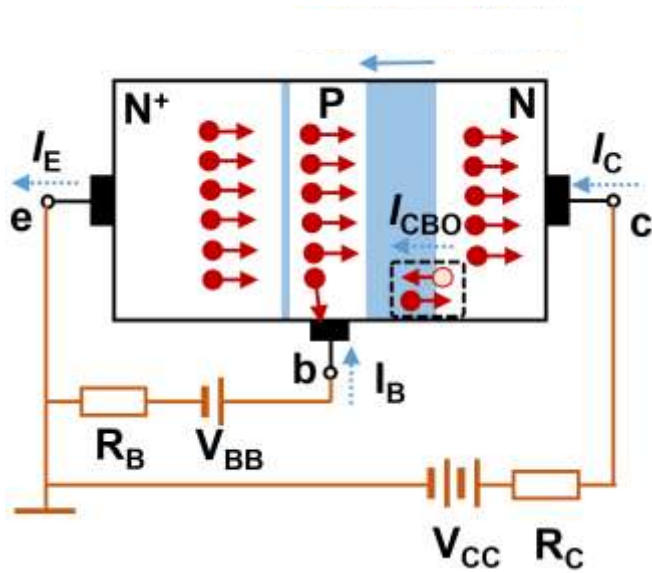
$$I_E \approx I_C \gg I_B$$

**Common-e DC
amplification factor**

共射极直流放大系数

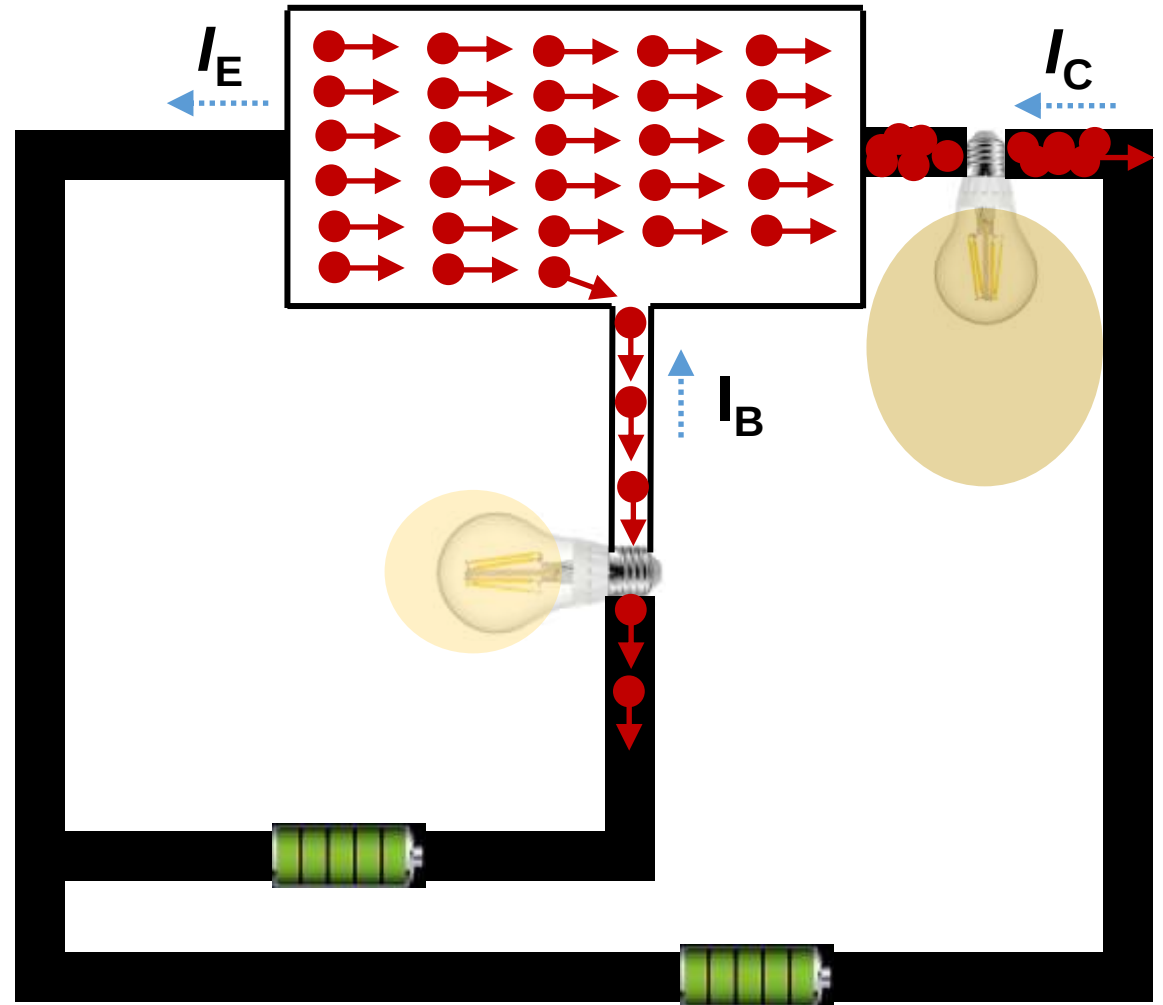
$$\bar{\beta} = \frac{I_C - I_{CBO}}{I_B + I_{CBO}} \approx \frac{I_C}{I_B}$$

$$\bar{\beta} = \frac{\bar{\alpha}}{1 - \bar{\alpha}} \quad \bar{\alpha} = \frac{\bar{\beta}}{1 + \bar{\beta}}$$



$$\bar{\beta} = \frac{I_C}{I_B} \gg 1$$

Common-emitter amplifier



Common-b DC amplification factor

$$\bar{\alpha} = \frac{I_C}{I_E}$$

$$\bar{\alpha} < 1, \bar{\alpha} \approx 1$$

Common-e DC amplification factor

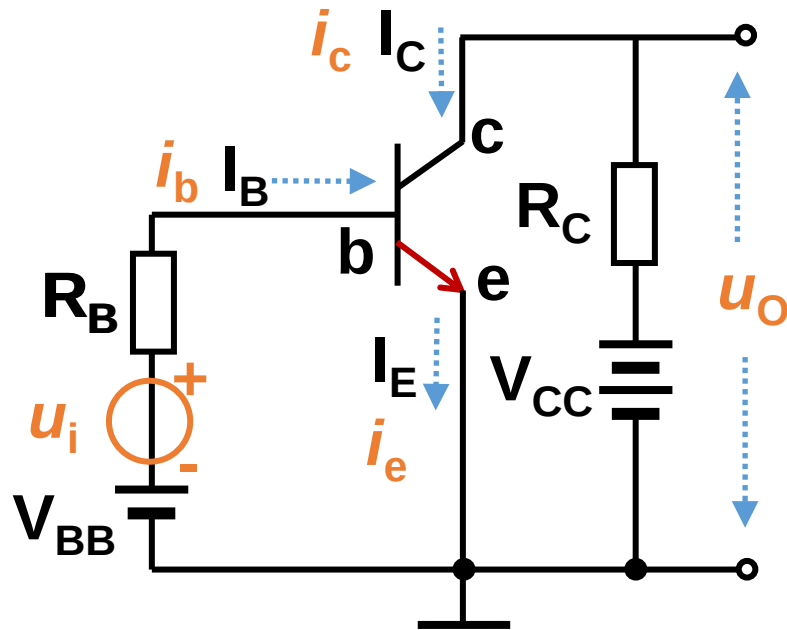
$$\bar{\beta} = \frac{I_C}{I_B}$$

$$\bar{\beta} \gg 1$$

$$\bar{\alpha} = \frac{\bar{\beta}}{1 + \bar{\beta}}$$

$$\bar{\beta} = \frac{\bar{\alpha}}{1 - \bar{\alpha}}$$

Common-emitter AC amplification circuit



Direct current

直流电流

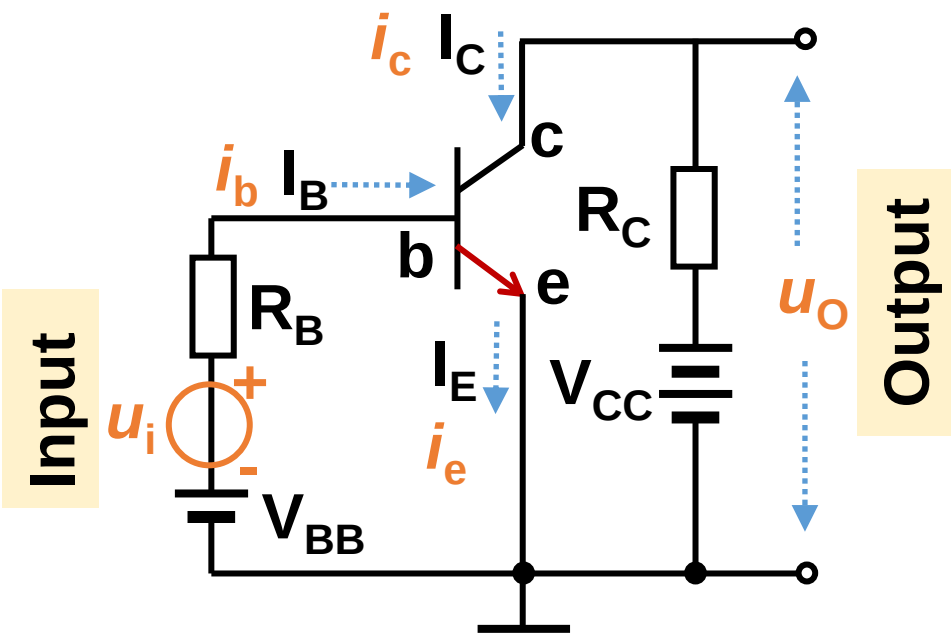
$$i_B = I_B + i_b$$

Total current
总瞬时电流

Alternating current
交流瞬时电流

$\dot{I}_b$: AC complex value 交流复数值

I_b : AC effective value 交流有效值



Input AC voltage signal u_i

$$i_B = I_B + i_b$$

$$i_C = I_C + i_c$$

Output voltage (DC+AC):

$$u_o = V_{CC} - i_c R_C$$

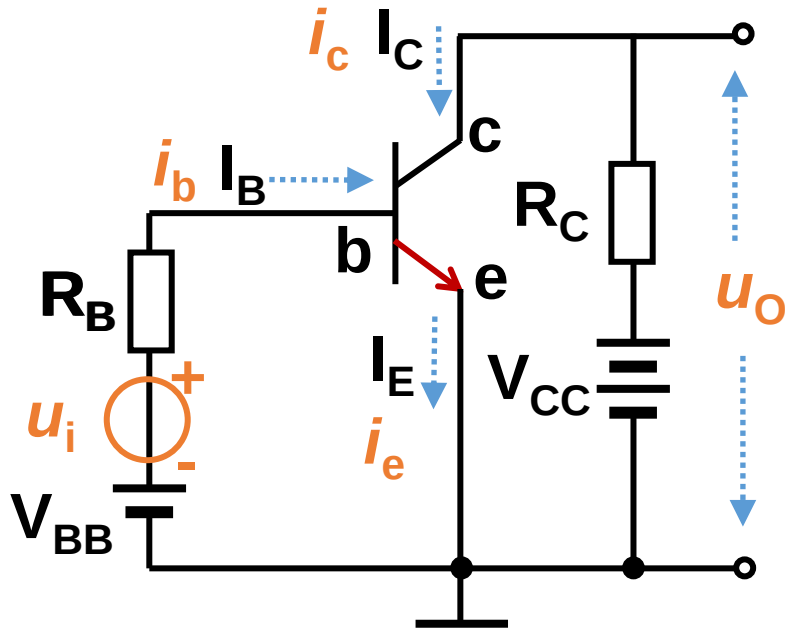
$$= V_{CC} - I_C R_C - i_c R_C$$

Output DC voltage: $U_O = V_{CC} - I_C R_C$

Output AC voltage: $u_o = -i_c R_C$

AC amplification factor

交流放大系数



Common-e: $\beta = \frac{i_c}{i_b}$?

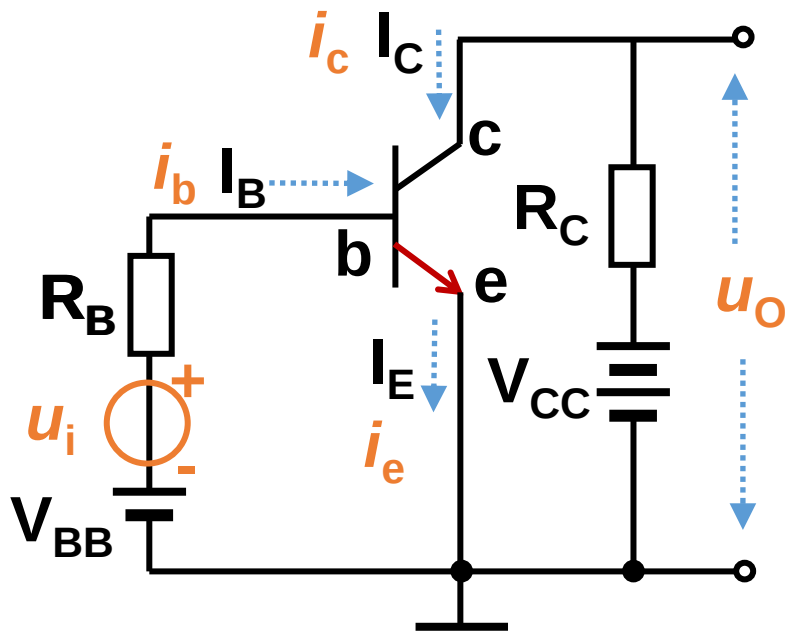
Common-b: $\alpha = \frac{i_c}{i_e}$?

In general case:

$$\alpha \approx \bar{\alpha} \quad \beta \approx \bar{\beta}$$

$$\beta = \frac{\alpha}{1 - \alpha} \quad \alpha = \frac{\beta}{1 + \beta}$$

AC amplification factor



Total current:

$$i_c = \beta i_B$$



$$I_C + i_c = \beta(I_B + i_b)$$



$$I_C = \beta I_B$$

$$i_c = \beta i_b$$

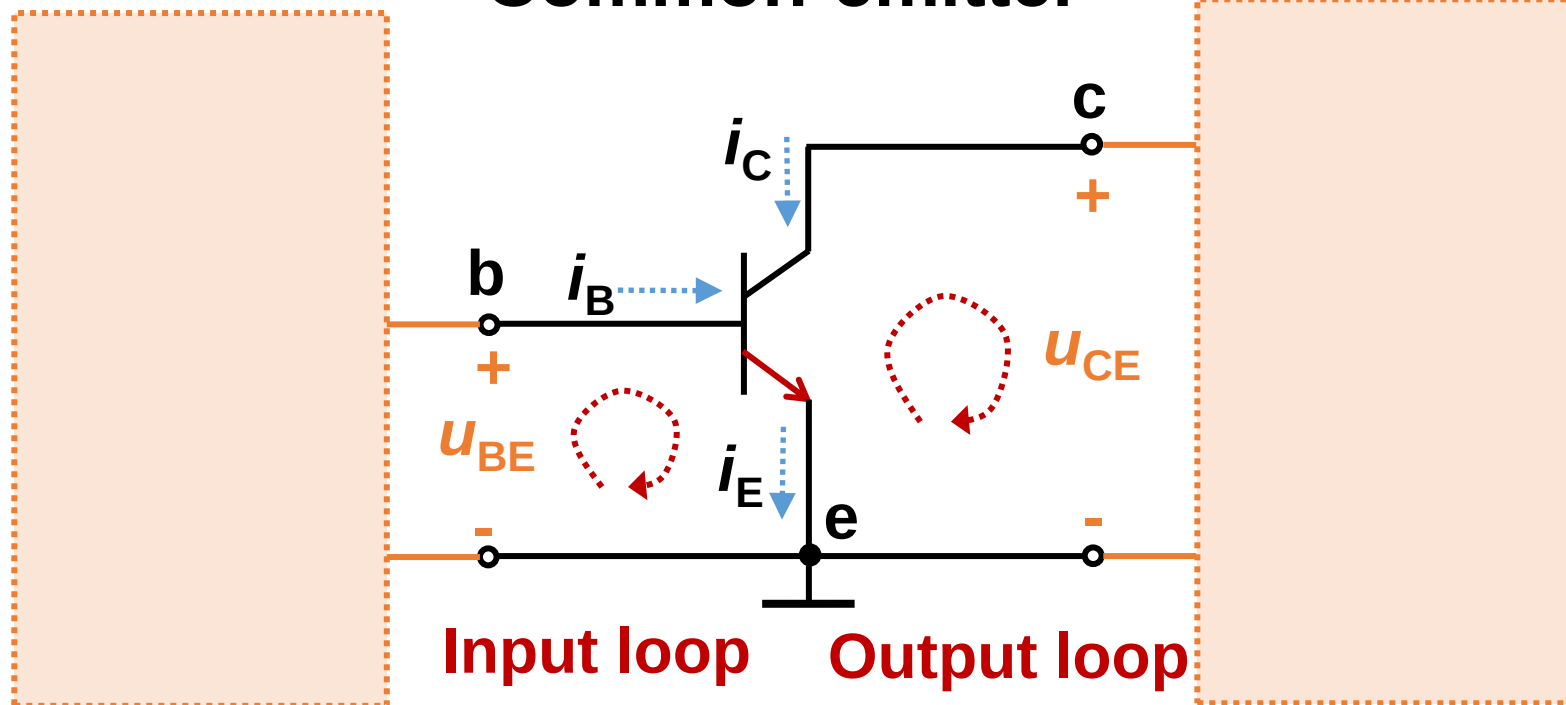


$$\beta = \bar{\beta}$$

$$\alpha = \bar{\alpha}$$

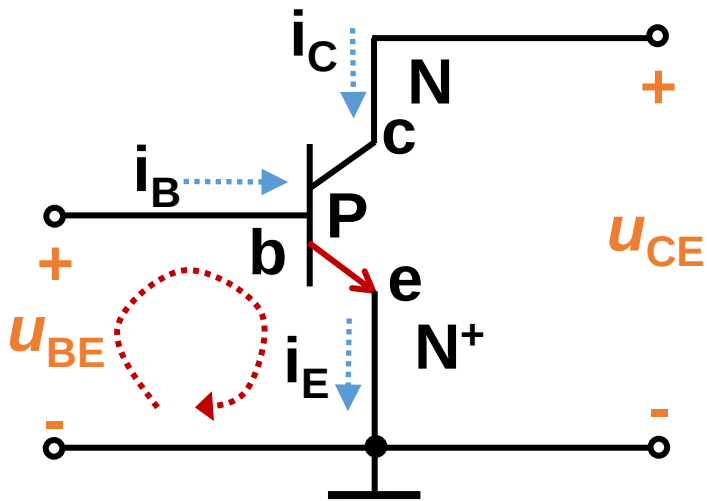
2.1.5 The input and output I-V characteristics

Common-emitter



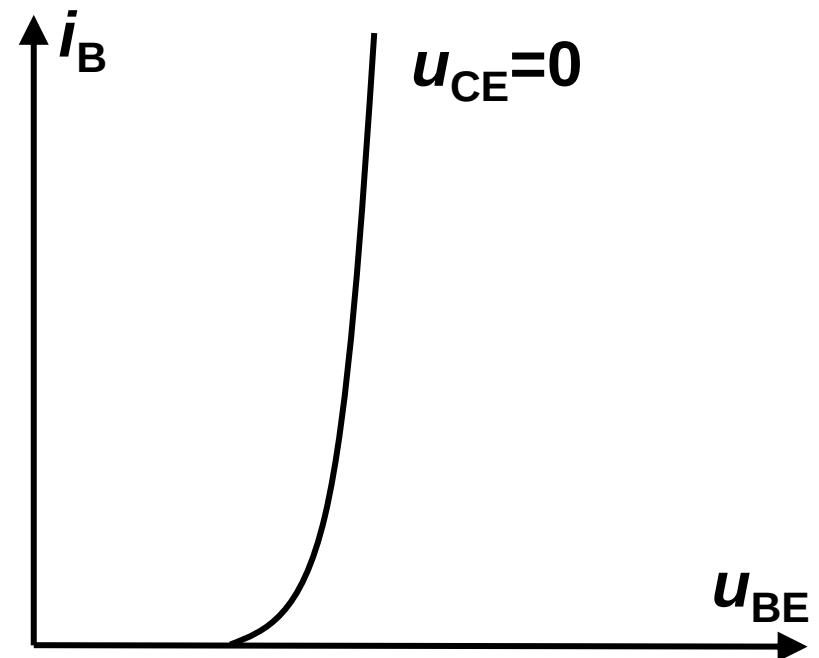
Input current: i_B
Input voltage: u_B

Output current: i_C
Output voltage: u_{CE}

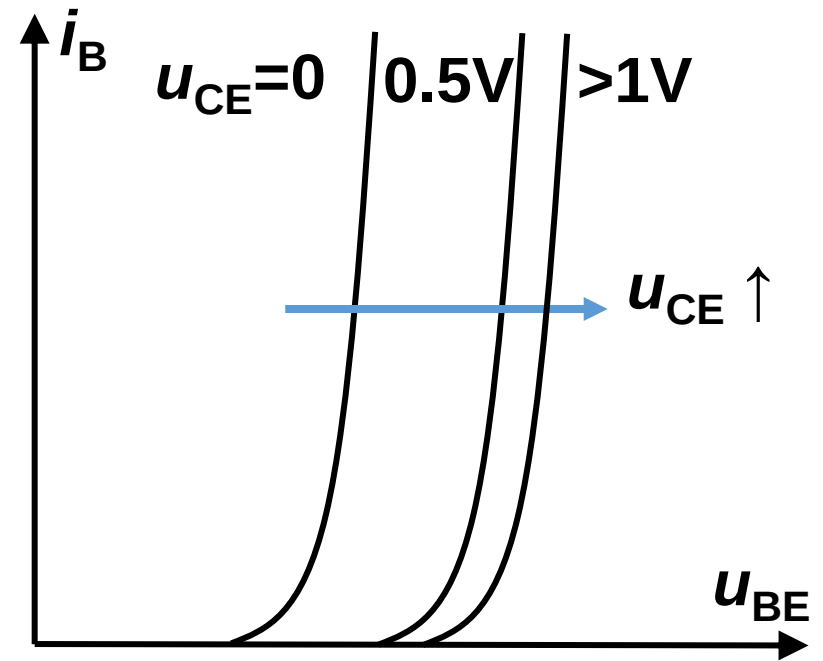
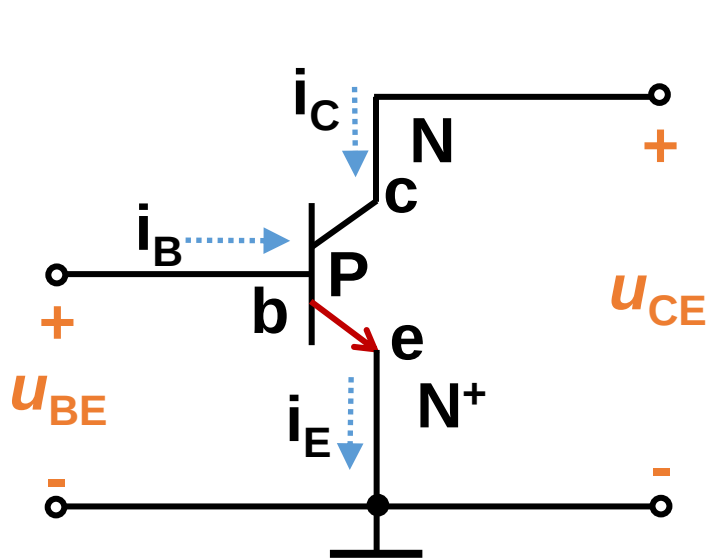


Input I-V characteristics:

$$i_B = f(u_{BE}) \Big|_{u_{CE} = \text{Constant}}$$

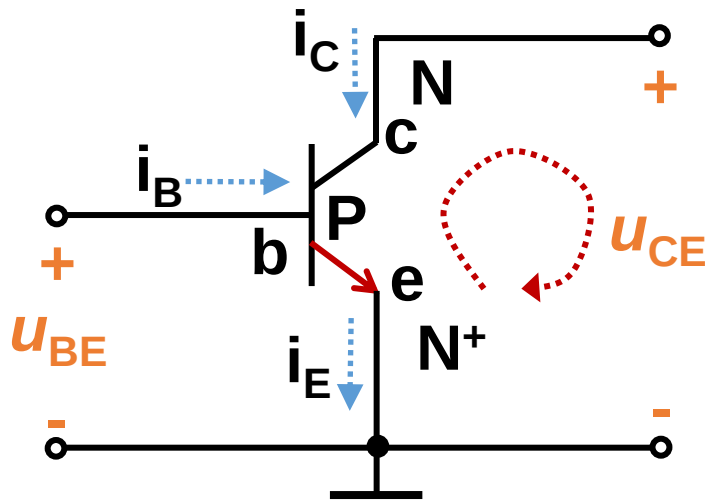


1) When $u_{CE}=0$, BE and BC junction are **both forward biased**. The internal electric field in BC junction is weak, most electrons are collected by the **Base**.



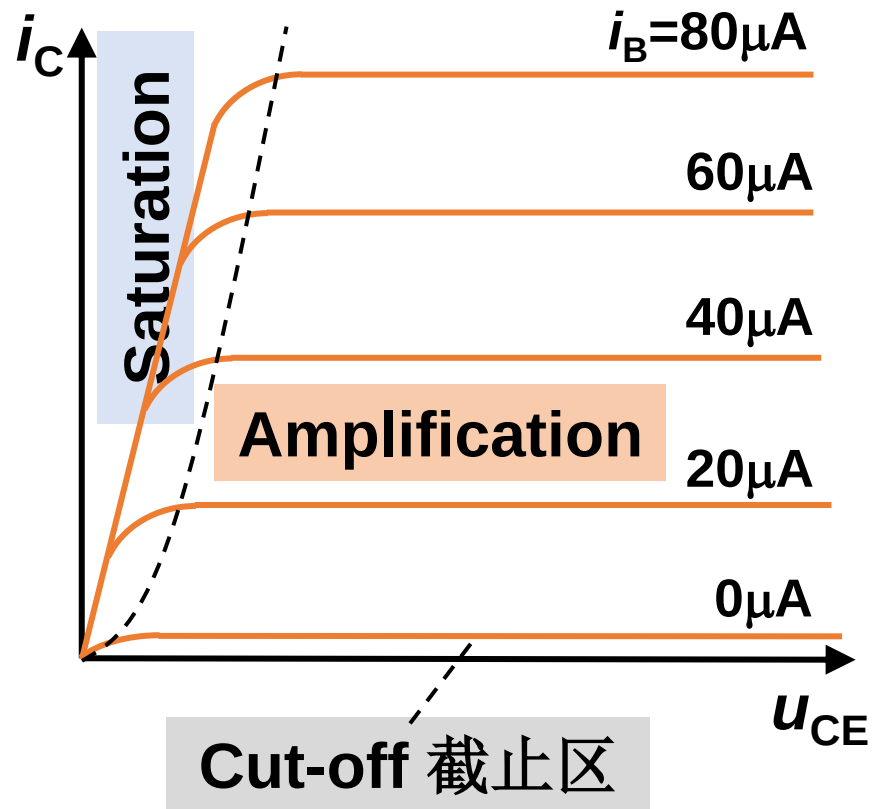
2) When u_{CE} increase ($<1V$), the electric field in BC junction increases. Some electrons are collected by **Collector**, and the left are collected by **Base**. To reach the same value of i_B , u_{BE} should be larger. Hence the I-V curve shifts to right.

3) When u_{CE} is large enough ($>1V$), the electric field in BC junction is strong enough, and most electrons are collected by **Collector** and **Collector** is saturated. Further increase u_{CE} , the curve does not shift any more.



Output I-V characteristics

$$i_C = f(u_{CE}) \Big|_{i_B = \text{Constant}}$$



1) Cut-off region:

$$i_B \approx 0$$



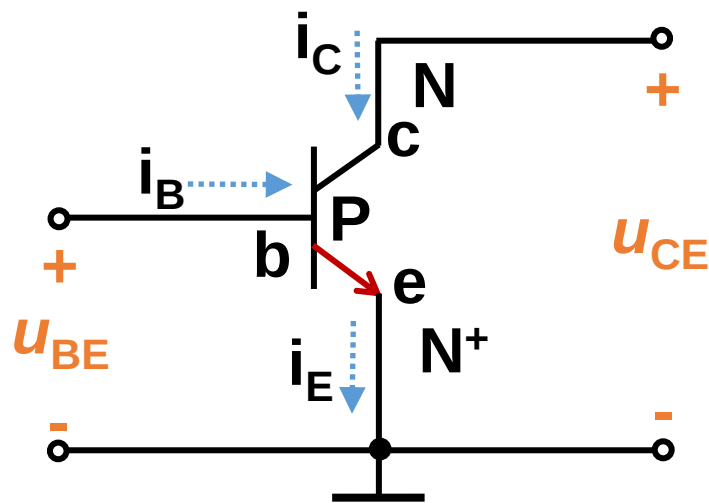
$$u_{BE} < U_{ON}$$

$$i_C \approx 0$$



$$u_{CE} > u_{BE}$$

U_{ON} : cut-in voltage of BE junction



2) Amplification region 放大区:

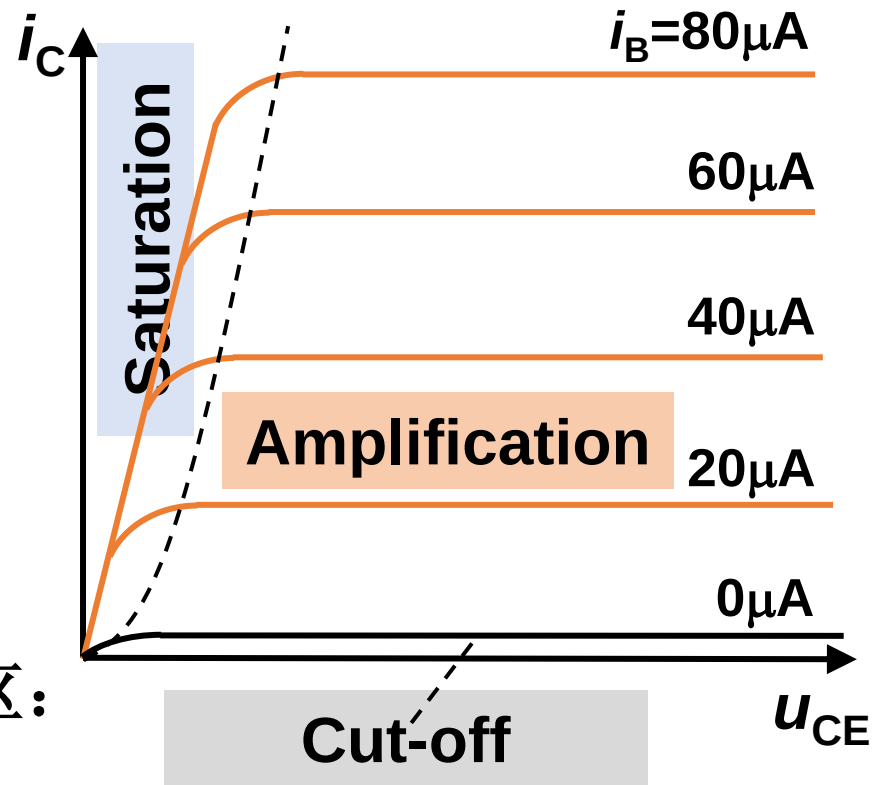
$$u_{BE} > U_{ON}$$

$$u_{CE} > u_{BE}$$

Si: $u_{BE} \sim 0.7 \text{ V}$; Ge: $u_{BE} \sim 0.2 \text{ V}$

BC junction is reverse biased

i_C is almost a constant



3) Saturation region 饱和区:

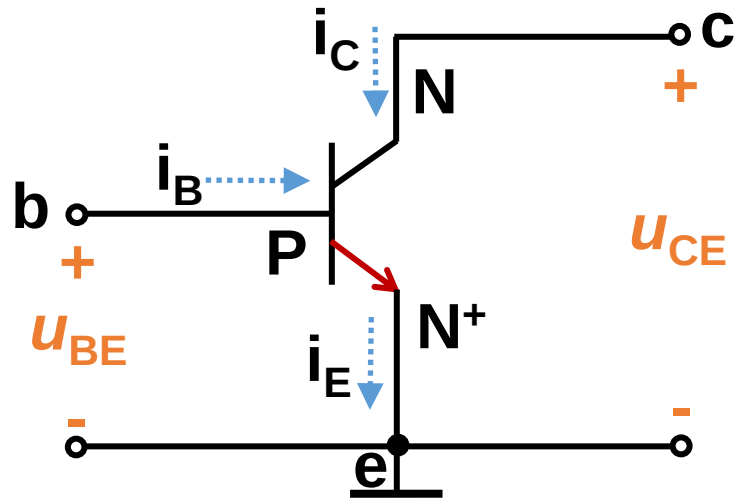
$$u_{BE} > U_{ON}$$

$$u_{CE} < u_{BE}$$

BC junction is forward biased

i_C increase with u_{CE}

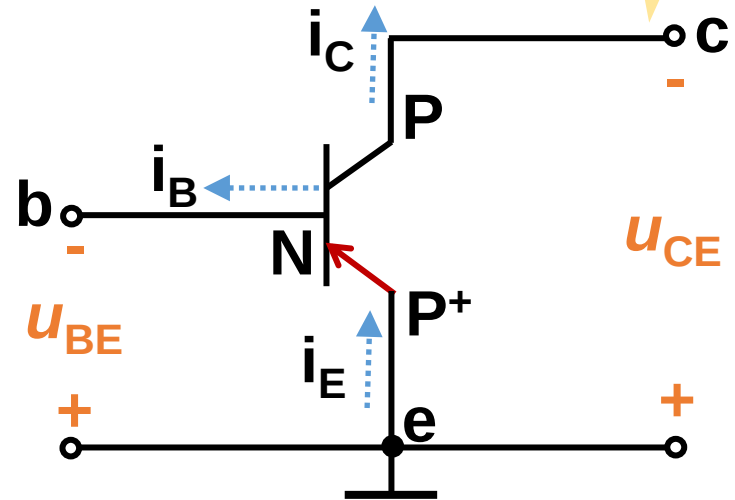
NPN



Amplification mode

PNP

Question!



Amplification mode

$$|u_{BE}| < |u_{CE}|$$

2.1.6 Limit parameters 极限参数

1) The reverse breakdown voltage of BE junction

$$U_{(BR)EBO}$$

2) The reverse breakdown voltage of BC junction

$$U_{(BR)CBO}$$

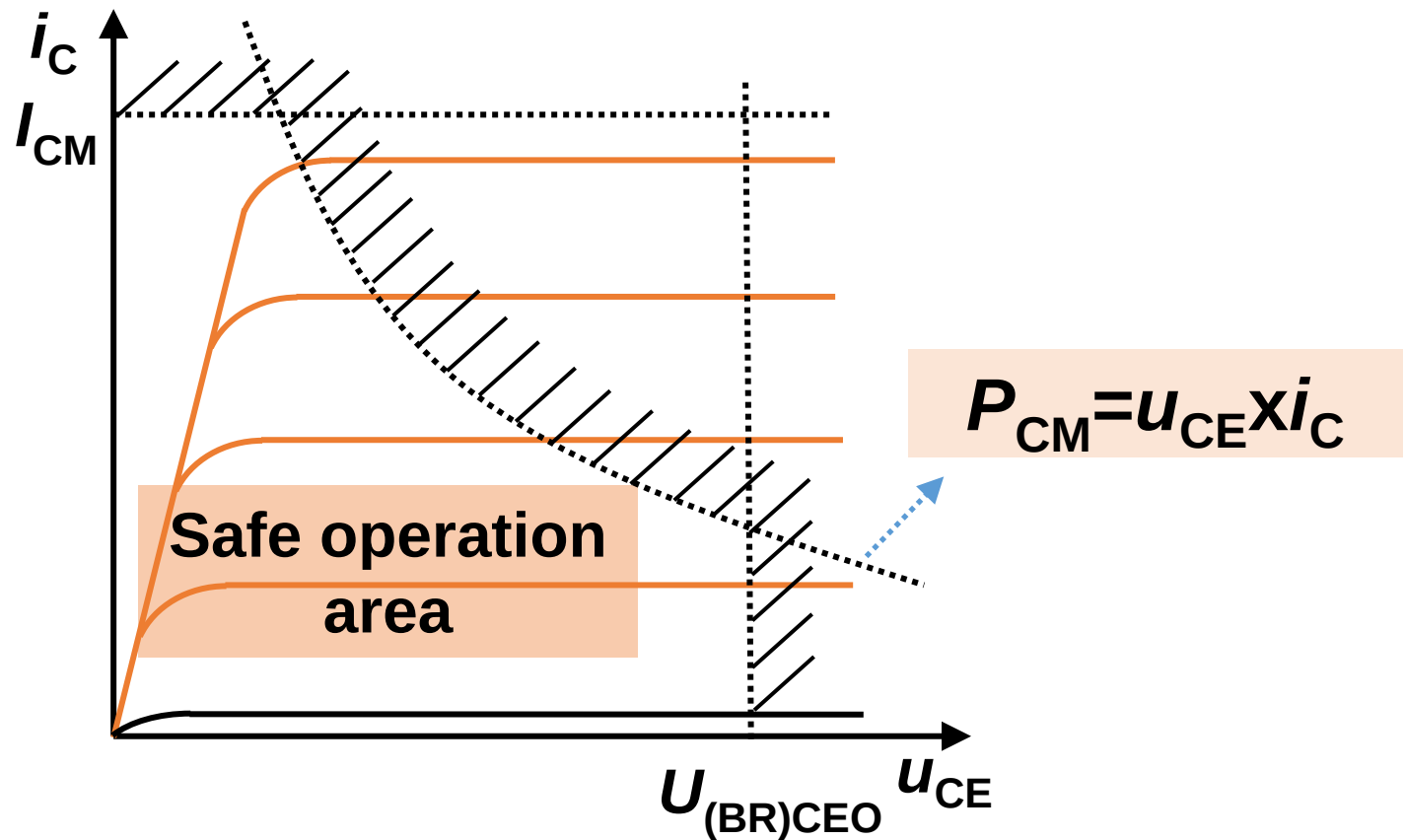
3) The reverse breakdown voltage of CE junction

$$U_{(BR)CEO}$$

4) Maximum current allowed in collector: I_{CM}

5) The maximum power dissipation allowed in collector: P_{CM}

P_{CM} determined the temperature of transistors. For Si transistor, $T < 150^\circ$. For Ge transistor, $T < 70^\circ$.



Homework 1-2: Refer to blackboard and QQ

Deadline: 2024-03-12 11:59pm

Diode	Cut-in voltage U_{ON}	Working voltage U_D	$U_D \approx$
Si	$\approx 0.5 \text{ V}$	0.6~0.8 V	0.7 V
Ge	$\approx 0.1 \text{ V}$	0.1~0.3 V	0.2 V

所有学生，包括重修的学生，都要提交。